

Selling Fast and Buying Slow: Heuristics and Trading Performance of Institutional Investors^{*}

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July 17, 2021

Are market experts prone to heuristics, and if so, do they transfer across closely related domains—buying and selling? We investigate this question using a unique dataset of institutional investors with portfolios averaging \$573 million. A striking finding emerges: while there is clear evidence of skill in buying, selling decisions underperform substantially—even relative to *random* selling strategies. This holds despite the similarity between the two decisions in frequency, substance and consequences for performance. Evidence suggests that an asymmetric allocation of cognitive resources such as attention can explain the discrepancy: we document a systematic, costly heuristic process when selling but not when buying.

Keywords: Heuristics, Behavioral Finance, Limited Attention, Expertise
JEL: D91

^{*}We thank Bronson Argyle, Ned Augenblick, Aislinn Bohren, Pedro Bordalo, John Campbell, Colin Camerer, Cary Frydman, Samuel Hartzmark, David Hirshleifer, Zwetelina Iliewa, Lawrence Jin, David Laibson, Ulrike Malmendier, Joshua Miller, Terry Odean, Alex Rees-Jones, Andrei Shleifer, Richard Thaler, Tuomo Vuolteenaho, Russ Wermers, conference and seminar participants at the NBER Behavioral Finance Meeting, Society of Judgment and Decision Making Conference, WFA Annual Conference, Booth School of Business, UC Berkeley, briq Institute, Caltech, ETH Zurich, Florida State, Harvard Business School, Higher School of Economics, New School of Economics, University of Amsterdam, and University of Frankfurt for helpful comments. We thank Kushan Tyagi, and especially Brice Green, for outstanding research assistance. The views expressed in the paper are solely those of the authors and do not necessarily represent the views of the institutions to which the authors are affiliated.

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Conflict-of-Interest Disclosure Statements

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I have reviewed the Disclosure Policy, and I have nothing to disclose.

Rick Di Mascio

I am disclosing that I am the Owner and CEO of Inalytics, the firm that has provided the data for the research paper. As such I have received income in excess of \$10,000 from Inalytics.

Inalytics' role has been to provide the data, and no Employee other than myself has been involved in the research.

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I have reviewed the Disclosure Policy, and I have nothing to disclose.

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I have reviewed the Disclosure Policy, and I have nothing to disclose.

1 Introduction

A large literature has demonstrated that market participants use heuristics and are prone to systematic biases. However, the majority of evidence comes from unsophisticated traders such as retail investors, who have been shown to be overconfident (Barber and Odean 2001), loss-averse (Larson, List, and Metcalfe 2016) and to exhibit limited attention in their trade decisions (Barber and Odean 2008; Hartzmark 2014). Comparatively little is known about the decision-making of market experts and it remains important to better understand whether they are prone to behavioral biases and, if so, the extent to which these biases affect performance.

This paper examines the decisions of sophisticated market participants—experienced institutional portfolio managers (PMs)—using a rich data set containing the entirety of their *daily* holdings and trades. Our data is comprised of 783 portfolios, with an average portfolio valued at approximately \$573 million. The modal PM is hired by a single institutional client such as a pension fund in a separately managed account to generate excess returns by forming concentrated portfolios that depart substantially from their benchmarks. More than 89 million fund-security-trading dates and 4.4 million high-stakes trades (2.0 and 2.4 million sells and buys, respectively) are observed between 2000 and 2016. We evaluate performance by constructing counterfactual portfolios, and compare PMs’ actual decisions to returns of the counterfactual strategy. Our data set uniquely allows us to evaluate selling decisions relative to a conservative counterfactual that assumes no skill: *randomly* selling an alternative position that was not traded on the same date.¹

We document a striking pattern: While the investors display clear skill in buying, their selling decisions underperform substantially. Positions added to the portfolio outperform both the benchmark and a strategy which randomly buys more shares of

¹Throughout the paper we construct counterfactuals based on *current* holdings. In a setting such as ours with active managers forming concentrated portfolios subject to short-sales constraints, evaluating a selling decision relative to a counterfactual which is unrelated to existing holdings (e.g., a benchmark index) is not an appropriate comparison. First, because portfolios depart quite a bit from the benchmark, selling the benchmark is often not feasible on the margin. Additionally, PMs may be selling in order to raise capital to buy, or because their opinion about a security has changed. An asset sold may outperform (underperform) a benchmark index, but the sale may be optimal (suboptimal) depending on what is bought with that capital and what other assets *could* have been sold (e.g. an alternative may have gone up even more).

assets already held in the portfolio by over 100 basis points annually per dollar of purchase volume. In contrast, selling decisions not only fail to beat a no-skill *random* selling strategy, they consistently underperform it by substantial amounts. In our preferred specification, PMs forgo 80 basis points per year relative to a factor neutral, random-selling strategy.² We perform a wide array of robustness checks, constructing counterfactuals that match assets bought and sold on size, value, idiosyncratic volatility, prior returns, and momentum, and ‘characteristic selectivity’ (Daniel, Grinblatt, Titman, and Wermers 1997). We also perform sample splits to account for the potential impact of outflow pressure and price impact. Our results are robust to these alternative specifications both in terms of statistical significance and economic magnitude. Heterogeneity analyses reveal that underperformance in selling appears most prominently amongst fundamentals-oriented managers who hold more active, concentrated portfolios with higher tracking error; PMs who rely on momentum strategies exhibit the least underperformance in selling. Finally, we show that the decomposed buying and selling measures have a meaningful relationship with overall portfolio returns: outperforming (underperforming) the counterfactual when buying (selling) is strongly predictive of higher (lower) excess portfolio returns.³

Why would a majority of portfolio managers appear to exhibit skill in buying while at the same time underperforming substantially in selling? At face value, the fundamentals of buying and selling to optimize performance are essentially the same: Both require incorporating information to forecast the distribution of future returns of an asset.⁴ Skill in both decisions requires the investor to look for relevant information and integrate it into the forecast. However, there is reason to suspect that selling and buying decisions involve different psychological processes (Barber and Odean 2013). Recent

²As a benchmark, active managers of mutual funds charge between 20 to 50 basis points per year in fees, depending on the size of the portfolio. The foregone returns due to poor selling are also substantially larger than the average cost differential (and net-of-fee performance differential) between mutual funds and institutional separately managed accounts (SMAs) of the sort we have in our sample, which is around 10-35 bp per annum (Chen, Chen, Johnson, and Sardarli 2017; Elton, Gruber, and Blake 2013).

³Moreover, we show that poor selling affects overall performance through two channels: the first is through the stocks sold outperforming the counterfactual (direct channel), and the second is through ‘fast’ selling leading PMs to prematurely discard viable investment ideas (indirect channel). As evidence for the latter, PMs rarely re-purchase assets that have left their portfolio, suggesting that sales eliminate equities from future consideration sets.

⁴As noted in Barber and Odean (2008), ‘in formal models, the decisions to buy and to sell often differ only by a minus sign.’

work from the lab is consistent with this discrepancy: Buying decisions appear to be more forward-looking and belief-driven than selling decisions in an experimental asset market (Grosshans, Langnickel, and Zeisberger 2018). Indeed, anecdotal evidence from our sample points to PMs thinking differently about and allocating different amounts of effort towards the two decisions; extensive interviews suggest that they appear to focus primarily on finding the next great idea to add to their portfolio and view selling largely as a way to raise cash for purchases.⁵

We argue that the stark discrepancy in performance between buys and sells is consistent with an asymmetric allocation of limited cognitive resources towards buying and away from selling. As a result of this asymmetry, we propose that while buying will resemble decisions made through standard portfolio optimization, selling decisions are driven by a two-stage process that has elements of bounded rationality entering at both stages. First, limited attention leads PMs to constrain their consideration set to assets with extreme attributes on a salient dimension—prior returns.⁶ Next, from this constrained consideration set, PMs choose to unload positions in which they have the least conviction. The latter effect can generate systematic underperformance if these positions happen to include neglected, but still viable investment ideas.⁷

As a first piece of evidence, we examine the performance trades that occur contemporaneously with exogenous events that draw investors’ attention to current holdings. The release of salient and portfolio-relevant information through company earnings announcements has been exploited to study limited attention in asset markets (Menkveld 2013). Earnings announcements, like other salient news releases, not only draw attention to specific assets or asset classes, they also provide new decision-relevant information (Ball and Brown 1968) on which skilled traders are able to capitalize (Ben-Rephael,

⁵The following quotes are illustrative of this attitude: “When I sell, I’m done with it. In fact, after I sell, I go through and delete the name of the position from the entire research universe.” “Selling is simply a cash raising exercise for the next buying idea.” “Buying is an investment decision, selling is something else.” In section 4.1, we also discuss why certain institutional features of our setting might lead PMs to prioritize buying decisions.

⁶Limited attention has been argued to generate a higher propensity to buy and sell assets with extreme returns (Hartzmark 2014; Ungeheuer 2017). Various measures of prior returns are among the most readily available pieces of information about a security – trading terminals and research platforms all highlight asset-specific prior returns – and research has shown that the salience of these return measures can affect investment decisions above and beyond the information they provide about future performance (Frydman and Rangel 2014; Frydman and Wang 2018).

⁷Barber and Odean (2008) argue for a similar two-stage trading process, writing that “preferences determine choices after attention has determined the choice set.”

Da, and Israelsen 2017; Easley, Engle, O'Hara, and Wu 2008; Fedyk 2018; Hendershott, Livdan, and Schürhoff 2015). We exploit variation in earnings announcements as predetermined shifters of attention which may lead PMs to think more deliberately about positions that they would have otherwise not considered selling. Accordingly, we predict that earnings announcements broaden the PMs' consideration sets and, as a result, contemporaneous selling decisions will outperform those made on non-announcement days. Since attention is already being channeled towards purchases, the performance of buying decisions will not change on announcement days. In contrast, if the difference in buying and selling performance is driven by some fundamental discrepancy between the two decisions, e.g. differences in skill, then trades should look similar on announcement and non-announcement days.

We find that selling decisions on earnings announcement days outperform those on non-announcement days by more than 150 basis points annually. Whereas sell decisions on non-announcement days substantially underperform (similar to the overall result), on average, stocks sold on announcement dates substantially *outperform* the random-sell counterfactual. Consistent with PMs focusing on buys throughout, we do not detect a systematic difference in performance of buying decisions on announcement versus non-announcement days. These results suggest that investors do not lack the fundamental skill to sell well—in fact, the point estimates of buying and selling performance on announcement days are similar—it is just not transferred.

As further evidence for the asymmetric allocation of cognitive resources, we document that PMs are prone to use a heuristic process when selling but not when buying. PMs in our sample have substantially greater propensities to sell positions that are extreme on the salient dimension of prior returns: both the worst and best performing assets in the portfolio are sold at rates more than *50 percent* higher than assets that just under- or over-performed. Non-psychological instrumental motives do not seem to explain this pattern: results are robust to controlling for position size and holding length, and are unlikely to be explained by risk management and tax motives. The pattern persists even after the inclusion of stock-date fixed effects which absorb a number of time-varying, stock-specific unobservables. On any given day, the *same* asset is more likely to be sold from a portfolio where it exhibits relatively extreme returns than from a portfolio where its recent performance stands out less compared to other

positions held. In contrast, we observe *no* similar tendency to focus on extremes on the buying side—unlike with selling, buying behavior correlates little with past returns.⁸ Prior returns appear to guide the PMs' consideration sets of what assets to sell but have little effect on decisions of what to buy.

Next, we show that given the consideration set of assets with extreme returns, PMs systematically choose to sell those that they have the least conviction in. We define 'conviction' as the extent to which the PM has developed a position as an integral, active part of her portfolio. This can be measured by examining the asset's weight relative to the benchmark: assets that score low on this dimension are most likely to be ideas that the PM began to develop but neglected to do so further. We find that these 'neglected ideas' are associated with much of the underperformance in selling; moreover, they are most likely to be sold and their sales exhibit the most pronounced relationship with extreme returns.⁹ In contrast, sales of 'high conviction' assets are not associated with any systematic underperformance.

Finally, we present evidence that the heuristic process outlined above is costly. Our analysis looks at within-manager variation in the propensity to sell assets with extreme returns. When the same manager becomes more likely to sell extremes (top quartile), they forgo almost 180 foregone basis points annually; in contrast, sales do not underperform when managers are least likely to sell extremes (bottom quartile). These results point to an empirical link between heuristic thinking and overall underperformance in selling. Moreover, we show that selling performance is further degraded during periods when attention devoted towards sales is likely to be stretched thin, such as when PMs are stressed (during periods when the overall portfolio is underwater) or selling in order to raise cash for buying decisions (attending to their selling choices even less). Importantly, the link between heuristic use and poor selling does not seem

⁸Since prior returns may reflect changes in relative valuations, it is not unreasonable to see a correlation between extreme prior returns and trading behavior. However, the revealed preferences of PMs' buying decisions suggest that the public signal provided by recent relative returns has little effect on PMs' beliefs about future expected returns. The lack of this correlation for buying decisions suggests that such instrumental motives are not the primary driver of behavior on the selling side.

⁹Note that the sale of assets with extreme returns may be sufficient to generate systematic underperformance if a large enough number of investors share assets that are categorized as extreme within their portfolios An (2015); An and Argyle (2016); Coval and Stafford (2007). However, as discussed in Section 2.1, this channel is unlikely to be driving the underperformance in selling within our sample as PMs tend to hold concentrated portfolios that depart quite a bit from the benchmark. To that end, we show that the correlation between assets sold in the cross-section is essentially zero.

to be driven by a persistent trait or lack of skill: consistent with the proposed attentional mechanism, differences in selling performance are associated with variation in covariates within-manager rather than in the cross-section.

Our paper contributes to the literature documenting biased decision-making in financial markets. The majority of research on investor behavior has focused on retail traders for whom daily holdings and trade data has been more readily available (see [Barber and Odean \(2011\)](#) for review). The selling pattern we document is most related to the rank effect described in [Hartzmark \(2014\)](#). There, retail investors appear to exhibit a similar pattern in selling *and* buying behavior—unloading and purchasing assets with more extreme returns. However, it is not clear from the data whether these trading strategies are costly: This set of investors have been found to underperform the market net of fees and display a host of heuristics and biases.¹⁰

While prior work has documented biases amongst experts in corporate finance settings, e.g. CEOs in charge of mergers ([Malmendier, Tate, and Yan 2011](#)) or other restructuring decisions ([Camerer and Malmendier 2007](#)), substantially less research exists on the behavioral biases of expert institutional investors. A number of papers have used data on aggregate returns to demonstrate slow and inefficient incorporation of certain types of signals into asset prices ([Chang, Hartzmark, Solomon, and Soltes 2016a](#); [Giglio and Shue 2014](#); [Hartzmark and Shue 2017](#); [Hong, Torous, and Valkanov 2007](#)). Although these findings highlight inefficiencies in the overall market, they cannot identify bias in expert investors per se. Other work has used the mandated release of quarterly holdings data to show the tendency of mutual funds to herd on the decisions of others ([Wermers 1999](#)), follow past prices ([Griffin, Harris, and Topaloglu 2003](#)), and display a ‘reverse’ disposition effect ([Chang, Solomon, and Westerfield 2016b](#)), but the coarseness of the data makes it difficult to identify biases from instrumental motives. As a result, the behavioral finance literature has mostly assumed unbiased institutional investors exploiting the behavioral biases of retail investors ([Malmendier 2018](#)). Our findings suggest that such an assumption may not be a valid one.

Our findings speak to the literature examining the performance of institutional investors. Although much of the research has argued that actively managed funds un-

¹⁰These biases include the disposition effect ([Odean 1998](#)), overconfidence ([Odean 1999](#)), and narrow bracketing ([Frydman, Hartzmark, and Solomon 2017](#)).

derperform the market after fees ([Gruber 1996](#); [Jensen 1968](#)), a number of studies have presented evidence for some skill in managers' ability to pick stocks (see [Wermers 2011](#), for a review). Using quarterly holdings data, [Wermers \(2000\)](#) shows that stocks held by mutual funds outperform the market on average, while net returns underperform. The majority of the difference can be explained by fees and transaction costs, suggesting skill in managers' ability to pick stocks. [Kosowski, Timmermann, Wermers, and White \(2006\)](#) employ a bootstrap analysis that identifies a sizable minority of funds who persistently beat the market even net of costs. [Puckett and Yan \(2011\)](#) argue that quarterly data may mask skill since interim trades (i.e. trades that are initiated and reversed within-quarter) can add considerable value to the fund. Finally, other work has looked at how fund characteristics are associated with performance; for example, [Chen, Hong, Huang, and Kubik \(2004\)](#) study the effect of fund size on returns and document a negative relationship.¹¹

Our results also complement the analysis of [Di Mascio, Lines, and Naik \(2017\)](#) and [von Beschwitz, Lunghi, and Schmidt \(2017\)](#), who used the Inalytics Ltd database.¹² They study the relation between purchase/sale volume (aggregated across PMs) and market conditions to test predictions of theoretical models of optimal strategic trading with private information. They find evidence that both opening and closing trades tend to earn positive risk-adjusted returns. Crucially, the analysis of closing trades in both papers does not consider feasible alternatives based on existing holdings. As such, while the papers can speak to skill in security selection, the question of whether or not PMs over- or under-perform with respect to feasible strategies and whether decisions are potentially subject to behavioral biases is outside the scope of their investigation.

Finally, our results also contribute to the literature demonstrating heuristics and biases amongst experts in domains such as sports ([Green and Daniels 2017](#); [Massey and Thaler 2013](#); [Pope and Schweitzer 2011](#); [Romer 2006](#)), judges ([Chen, Moskowitz, and Shue 2016](#)), professional forecasters ([Coibion and Gorodnichenko 2015](#)), monitoring of corporate actions ([Kempf, Manconi, and Spalt 2016](#)), and retail markets ([DellaVigna and Gentzkow 2017](#)). This line of work highlights the persistence of behavioral biases

¹¹Papers have also looked at how the presence of institutional investors affects prices ([Gompers and Metrick 2001](#)), market stability ([Brunnermeier and Nagel 2004](#)), and corporate governance ([Ferreira and Matos 2008](#)).

¹²Whereas the former also studies a sample of long only portfolios similar to ours, the latter restricts attention to a smaller set of 21 long-short hedge funds.

despite significant experience and exposure to market forces.

The paper proceeds as follows. Section 2 describes the data. Section 3 presents results on performance of buying and selling decisions, while Sections 4 and 5 present results on the use of heuristics in trading strategies and how those strategies affect performance. Section 6 concludes.

2 Institutional Sample

This section reviews the data sources which are assembled for our analysis and presents some relevant background and summary statistics on the data we observe in our sample of institutional portfolios. Further details are in Appendix sections A.1 and A.2.

2.1 Empirical Setting: Institutional Portfolio Managers

Our primary source of data is compiled by Inalytics Ltd. These data include information on the portfolio holdings and trading activities of institutional investors. Inalytics acquires this information as part of one of its major lines of business, which is to offer portfolio monitoring services for institutional investors that analyze the investment decisions of portfolio managers.¹³ The majority of portfolios in our sample are sourced from asset owners—institutional investors such as pension funds who provide capital to PMs to allocate on their behalf. In these cases, we see holdings and trades related to the specific assets owned by the client. The remainder of the portfolios are submitted by PMs themselves who seek to benchmark their own performance; in these cases, data will frequently correspond with holdings and trades aggregated over multiple clients. These data are associated with a single strategy, so we do not observe assets managed by the same PMs using alternative strategies.

For purposes of this study, Inalytics assembled a dataset of long-only equity portfolios spanning from January 2000 through March of 2016. These portfolios are almost always tax-exempt, hold limited cash, and are prohibited from using leverage or taking short positions. The portfolios are internationally diversified, including data from a large number of global equity markets. Data are only available during periods for which Inalytics' monitoring service is performed. After applying a sequence of filters

¹³We will use the terms fund and portfolio interchangeably throughout our discussion.

described further in Appendix A.1, our final sample includes daily information about holdings and trades covering about 51 thousand portfolio-months of data, which are compiled from a set of 783 institutional portfolios. During this period, we have an average of just over 5 years (65 months) of data per portfolio and observe 89 million fund-security-trading date observations, 2.4 million buy trades, and 2 million sell trades. We convert all market values to US dollars at the end of each trading day.¹⁴

Our sample consists entirely of active managers, and, based on our review of companies' publicly available marketing materials, the vast majority of these PMs identify mispriced securities through a mix of balance-sheet analysis, conversations with management, and qualitative judgements. There are only 22 systematic, quantitative portfolios in our sample, and our results are insensitive to excluding them. Section A.2 in the Appendix provides a summary of several qualitative characteristics of our sample. Portfolios are mostly beholden to a single specific institutional client—the modal portfolio is managed by a PM on behalf of a pension fund.

Relative to the well-studied universe of mutual funds, the portfolios in our sample are highly concentrated and have large tracking error budgets.¹⁵ These types of high-conviction PMs are typically hired by pension funds or endowments to generate alpha as a complement to other managers with more passive strategies. As a consequence, the PMs in our sample are expected to be concentrated stock-pickers with low correlation to other assets. Goyal, Wahal, and Yavuz (2020) describe the client's search process for managers in detail, where an institutional plan sponsor puts out a request for proposals, hears presentations from three to five finalists, and finally selects a manager from amongst this group. The authors find two important factors predict which managers are selected: past returns and previous interactions.¹⁶ Speaking loosely, the selection of highly active managers requires a record of high past returns and favorable qualitative judgements about their quality.¹⁷

¹⁴We compile data on exchange rates from three sources: Datastream, Compustat Global, and Inalytics' internal database, with Datastream being our primary source. In the vast majority of cases, at least two of these sources have identical exchange rates.

¹⁵In results presented in Tables 3, A.4, and A.5, we show that tracking error is not likely to be a binding constraint.

¹⁶Yale, for example, invites managers for a weekend of tennis and golf to discuss investment process and philosophy (<https://www.institutionalinvestor.com/article/b1k52pp0nq4f8b/The-Easily-Misunderstood-Yale-Model>).

¹⁷The CFA Institute makes a similar claim, saying that selecting a manager is based primarily on, "a

This institutional arrangement has several important consequences. First, because the sample is made up almost entirely of institutional separately managed accounts (SMAs), the PMs are subject an entirely different flow structure compared to mutual funds. Mutual funds invest in a common set of securities on behalf of all of their shareholders, each of whom has essentially been promised daily liquidity. In contrast, managers of an SMA are typically beholden to a single client, and as a result, direct outflows are unlikely to cause forced sales. Flows in or out of portfolios tend to either be small withdrawals for cash flow needs of a pension, which are entirely predictable by the PMs, or large block withdrawals that come from firing a manager. In these latter cases, asset owners engage third party transition managers to assist with transfer of assets to new PMs. As a result, the prospect of unexpected changes in assets under management (AUM) is unlikely to significantly affect PMs' decision-making and flow-induced sales tend to be rare (relative to mutual funds).¹⁸

2.2 Descriptive Statistics

For each portfolio, we have a complete history of holdings and trades at the daily level throughout the sample period. Inalytics collects portfolio data on a monthly basis and extends them to a daily basis by adjusting quantities using daily trades data. As a result, we observe the *complete* equity holdings of the portfolio at the end of each trading day (quantities, prices, and securities held), as well as a daily record of buy and sell trades (quantities bought/sold and prices) and daily portfolio returns, though we do not observe cash balances. Each portfolio is associated with a specific benchmark (usually a broad market index) against which its performance is evaluated. Our dataset includes an unbalanced panel of both active and inactive portfolios, with the vast majority of the data collected in real-time, suggesting that incubation and survivorship biases are not a substantial concern for our analysis.¹⁹

quantitative analysis of the manager's performance track record, and a qualitative analysis of the manager's investment process."

¹⁸Our results support this conjecture: we show that buying and selling performance is not qualitatively affected by proxies for flows. This is in contrast to the results of [Alexander, Cici, and Gibson \(2007\)](#) who report that flow-induced buys and sells substantially underperform discretionary trades in a sample of mutual funds. See also [Chen et al. \(2017\)](#); [Edelen \(1999\)](#).

¹⁹Furthermore, given that the majority of our analyses involve comparisons of stocks held with stocks traded, a number of common portfolio-specific factors which could potentially be associated with incubation/survivorship biases are differenced out via our methodology.

Table 1. Portfolio level summary statistics

This table reports summary statistics for various monthly variables from the analysis dataset of 783 portfolios. See Appendix Table A.2, Appendix A.1-A.2, and main text for additional details on variable construction and summary statistics.

Variable	Count	Mean	Std	25th	50th	75th
Assets under management (\$million)	51228	573.6	1169.3	71.70	201.8	499.0
Number of stocks	51229	78.49	68.46	40.95	58.60	86.58
Turnover(%)	51223	4.10	5.76	0.93	2.54	5.03
Fraction of distinct stocks sold over all holdings (%)	51221	10.14	12.13	1.923	5.695	13.70
Fraction of distinct stocks bought over all holdings (%)	51221	14.86	17.68	3.788	8.820	19.23
Monthly benchmark-adjusted returns (%)	48786	0.22	1.77	-0.60	0.17	1.01
SD of daily benchmark-adjusted returns (%)	48041	0.35	0.21	0.21	0.29	0.43
Loading on Market	48705	0.97	0.26	0.81	0.94	1.12
Loading on SMB	48705	0.01	0.50	-0.32	-0.06	0.27
Loading on HML	48705	-0.06	0.50	-0.36	-0.07	0.22
Loading on Momentum	48705	0.05	0.34	-0.13	0.04	0.22

To complement these data we merge in external information on past and future returns (including periods before and/or after we have portfolio data). When possible, we use external price and return series from CRSP; otherwise, we use price data from Datastream. When neither of these sources are available, Inalytics provided us with the remaining price series which are sourced (in order of priority) from MSCI Inc. and the portfolio managers themselves. Since PMs in our sample have global mandates and tend to hold concentrated portfolios that depart quite a bit from the benchmark, there is a fairly low overlap between stocks held contemporaneously by different portfolios in our sample.

Using these data we construct a wide array of measures at the portfolio-time and portfolio-stock-time (position) level. Table 1 summarizes a number of key monthly fund characteristics. For brevity, we emphasize a few key attributes of our sample here and relegate discussion of summary statistics for many of these variables to Appendix Table A.2 and section A.2, respectively. All portfolios are large, and there is considerable heterogeneity in portfolio size.²⁰ In addition, funds differ noticeably in terms of their

²⁰Note that given the size of the portfolios (average AUM \$533 million), trades in the 1 to 5 million dollar range are unlikely to face issues related to price impact, such as difficulty exiting from a position. These magnitudes are in sharp contrast with trades of big mutual funds (e.g. between \$25 and \$200 billion dollars) which have been examined in prior work (Coval and Stafford 2007; Lou

trading activity levels. Average monthly turnover is about 4 percent of assets under management, but some funds are considerably more active in their trading behavior than others (the standard deviation is 5.7 percent).

While holding fairly diversified portfolios (average number of stocks is about 78 with a standard deviation of 68), funds in our sample remain active, with positions that deviate substantially from their benchmarks. This is more concentrated than the average mutual fund, which holds a median of 92 stocks during our sample period [Wermers, Yao, and Zhao \(2012\)](#). The average tracking error—the standard deviation of the difference between the daily portfolio return and the benchmark—is about 0.35 percent per day, or about 5.7 percent on an annualized basis. As discussed in the previous subsection, this corresponds to portfolios that are substantially more concentrated and have larger tracking error budgets than the typical passive or retail mutual fund. On average, a manager will initiate a sell trade for about 10 percent and a buy trade for about 15 percent of the stocks in his/her portfolio each month. We also characterize fund portfolios in terms of factor exposures by computing rolling Carhart 4-factor regressions (using the prior 1 year of daily data with the Fama-French international factors), adjusted for asynchronous trading.²¹ The average market beta is about 1, and average exposures to the SMB, HML, and Momentum factors are fairly close to zero.

The average fund in our sample beats its respective benchmark by about 0.22 percent per month, or 2.6 percent per year. This, in conjunction with the fact that funds' average betas are close to 1 and have little average exposure to the three other priced risk factors, suggests that these managers are highly skilled.²² We view the positive selection of managers in our sample as an advantage when studying expertise and heuristic use: The population we examine is clearly skilled, and thus identifying biased behavior is likely a lower bound when generalizing the results.

In Appendix [A.2](#), we summarize a variety of additional attributes of our sample. Relative to US mutual funds, PMs in our sample have a larger share of international holdings. Managers have concentrated holdings and long holding periods. Specifically,

[2012](#)).

²¹Following [Dimson \(1979\)](#), we adjust for asynchronicity by including one lag and one forward return of each factor. We use these adjustments throughout when estimating daily factor loadings.

²²Prior work has demonstrated that a subset of institutional investors do persistently outperform the market and generate alpha ([Kosowski et al. 2006](#)).

the average holding length is at least 485 calendar days (or about 15 months).²³

Differences from other datasets This sample offers some unique opportunities to study expert decision-making relative to other datasets in the literature. First, in contrast to the Large Discount Brokerage dataset of [Barber and Odean \(2000\)](#), which features portfolio holdings and trades of individual retail investors, our data include complete portfolio and trade-level detail for a population of professional investors managing large pools of assets.²⁴ Illustrative of this distinction, [Barber and Odean \(2000\)](#) report that the value of the average portfolio is \$26,000 and that the *top quintile* of investors by wealth had account sizes around \$150,000—the average portfolio in our sample is almost four *thousand* times larger. Second, unlike other datasets which characterize institutional portfolios such as mutual fund portfolio holdings reports and 13-F filings (e.g. [Chang et al. \(2016b\)](#)), we are able to observe portfolio holdings and changes to those holdings on a *daily* level. This allows us to test hypotheses on individual decision-making that are infeasible with quarterly data. Additionally, in the other most widely used database with institutional trading information—the Abel Noser/ANcerno database (for an overview, see [Hu, Jo, Wang, and Xie 2018](#))—researchers often do not observe all trades made by a given institutional investor and tend to lack timely information on portfolio holdings. Finally, unlike passive and retail mutual funds, our sample is comprised of professional managers who are mostly beholden to a single client (e.g. pension fund) and are paid to generate alpha through concentrated portfolios that depart from the respective benchmark. These differences are highlighted in Appendix [A.2](#), which compares characteristics of portfolios in our sample to those in other datasets. As discussed further in Section [3](#), one notable consequence of this is that PMs are not very constrained by client-mandated flows or tight tracking error budgets (in contrast to mutual funds, e.g. [Alexander et al. \(2007\)](#) or earlier studies of pension funds, e.g. [Del Guercio and Tkac \(2002\)](#)). This suggests that trading decisions are largely up to the managers' discretions.

²³This is an underestimate: our measure is right censored since we can only measure holding length beginning with the first portfolio snapshot.

²⁴See [Barber and Odean \(2011\)](#) for a survey of studies using this and other similar datasets.

3 Overall Trading Performance

Having described the basic properties of our dataset, we now examine performance of PMs' decisions. We begin by discussing our methodology for computing counterfactual portfolio returns and, accordingly, value-added measures. We then present the first of our empirical results, which correspond to the average value-added (or lost) associated with managers' active buying and selling decisions.

3.1 Constructing counterfactuals

Given that PMs in our sample tend to hold limited cash positions and are not generally permitted to use leverage, the primary mechanism for raising money to purchase new assets is selling existing ones. Since the portfolios already include stocks that are carefully selected to outperform their respective benchmarks, the choice of which asset to sell is far from innocuous. Precisely if managers' use of information makes them skilled at picking stocks, biased selling strategies have the potential to cannibalize existing, still viable investment ideas and to reduce the potential value for executing new ones. It is therefore important to construct the appropriate benchmark to serve as the counterfactual for evaluating buying and selling decisions. In contrast, we would expect unskilled investors neither to gain nor lose money (on a risk-adjusted basis) by relying on a simple rule of thumb for selling existing positions.

The fact that we observe daily transactions allows us to compare observed buy and sell decisions to counterfactual strategies constructed using concurrent portfolio holdings data. Our measures correspond to the relative payoffs from two hypothetical experiments: one for evaluating buying decisions, and one for evaluating selling decisions. For evaluating buys, suppose that we learned that a manager was planning to invest \$1 to purchase a stock tomorrow and to hold it for a fixed period of time. We then suggest that instead of executing the proposed idea, the PM invests that money in a randomly selected stock from her other holdings. Likewise, we can suggest that instead of selling a particular stock, the PM randomly sells one of her other positions to raise the same amount of cash, holding the stock that was to be sold for the same period.

Since our conditioning information was also available to the manager and our strate-

gies are always feasible on the margin, one would expect decisions of a skilled PM to outperform our counterfactual.²⁵ Note that the expected payoff from the counterfactual strategy (integrating out uncertainty about which stock is randomly selected) corresponds to the equal-weighted mean of realized returns across stocks held in the portfolio, which we denote by R_{hold} . Similar results obtain if we use lagged portfolio weights to construct a value-weighted mean instead. The manager's decision adds value relative to the random counterfactual if $R_{buy} - R_{hold} > 0$ in the first experiment and if $R_{hold} - R_{sell} > 0$ in the second experiment. Following this logic, we compute $R_{buy} - R_{hold}$ and $R_{hold} - R_{sell}$ over horizons ranging from 1 week to 1 year for all buy and sell trades, respectively, to characterize the value-added associated with each.

If the return measure of interest is a cumulative return over the relevant horizon, these measures capture the impact on benchmark-adjusted returns associated with switching from the counterfactual to the actual trade, per dollar transacted. According to our discussions with clients and managers these relative returns are the primary metric by which our PMs are evaluated.²⁶ That said, given that stocks should earn ex-ante compensation for systematic risk exposures, our preferred measures will use “factor-neutral” stock returns, all of which should earn the same expected return per period. We estimate stock-level exposures to the Fama-French/Carhart 4 factors using pre-trade data, then use them to adjust stock-level returns to hedge ex-ante differences in exposures. Further details about this method are presented in Appendix A.3.

We aggregate across trades and conduct inference as follows. If multiple stocks are bought or sold on a given day, we average these measures for buy and sell trades separately. Since not all funds trade every day and are not necessarily present throughout our sample period, this averaging procedure yields a portfolio-day unbalanced panel. Because some funds trade much more frequently than others—see the dispersion in

²⁵In contrast, selling the benchmark to finance a purchase, which implicitly corresponds to the counterfactual in measuring benchmark-adjusted returns of stocks sold, is likely infeasible for a long-only manager who, similar those in our sample, holds a portfolio with a small (relative to the number of assets in the benchmark) number of high active share positions and thus deviates substantially from the benchmark. Purchasing the benchmark is feasible on the other hand. We discuss concerns about transaction costs and price impact below.

²⁶These measures also have an alternative interpretation to the extent that buy and sell trades are not motivated by a desire to change a portfolio's systematic risk exposures. In that case, we would expect loadings on priced factors of the assets being traded and the hold portfolio to be similar and these measures would also correspond to differences in risk-adjusted returns (i.e., “alpha”). Generally, we find similar results regardless of our approach for correcting for systematic risk.

monthly turnover in Table A.1—we weight observations inversely to a measure of trading frequency.²⁷ We compute double-clustered standard errors using a panel estimator similar to Hansen and Hodrick (1980). It allows individual fund time series to be serially correlated and additionally allows these value-added measures to cross-sectionally correlated across PMs who place trades at similar times. While the autocorrelation correction is likely required since our use of long horizon returns potentially introduces an overlapping structure in the error term of each fund’s value-added time series, the latter adjustment turns out to be small. For further details, see Appendix A.4.

3.2 Overall performance relative to counterfactuals

Figure 1, Panel A shows average factor neutral counterfactual returns for buying decisions. As will turn out to be the case across all of our specifications, we find strong evidence that purchases add value relative to the random buy counterfactual, $R_{buy} - R_{hold}$. The average stock bought outperforms the counterfactual by nearly 120 basis points over a one year horizon.

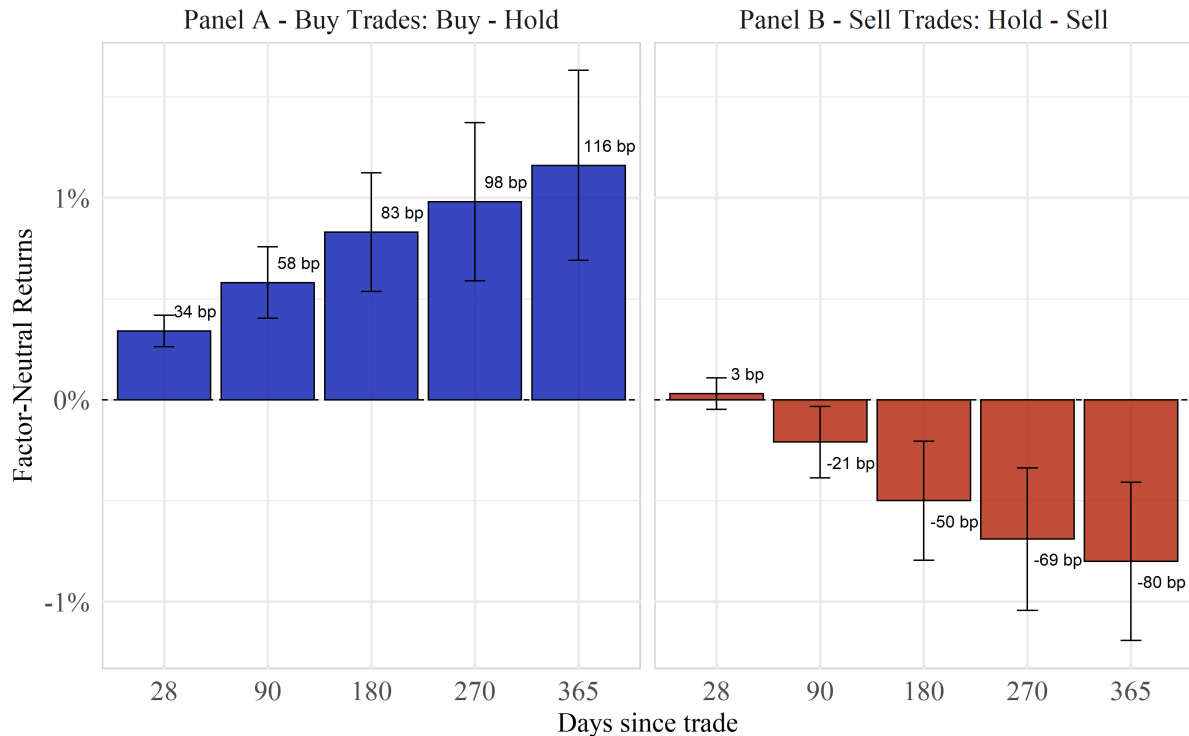
Figure 1, Panel B presents average value-added, $R_{hold} - R_{sell}$, for sell trades relative to a factor neutral counterfactual. Recall that our measure is already signed so that positive values indicate that a trade helps portfolio performance relative to the counterfactual, and negative values point to a trade hurting performance. In stark contrast to Panel A, these estimates suggest that managers’ actual sell trades *underperform* a simple random selling strategy. Magnitudes are quite substantial: The value lost from an average sell trade is indistinguishable from zero at a 1 month horizon but on the order of 80 basis points at a 1 year horizon relative to a simple counterfactual which randomly sells other stocks held on the same day.

To help assess magnitudes associated with these estimates, Table 2 links our performance measures with portfolios’ benchmark-adjusted returns. Specifically, Inalytics provides a daily estimate of the benchmark-adjusted return on each portfolio, and, for each portfolio, we compute the average of this return compounded over the next

²⁷We weight observations inversely to the number of trading days in a calendar year that the fund buys and sells a stock. This measure allows for an easier comparison across buys and sells, since we use the same weights across both types of trades. We obtain similar results when we instead weight inversely to the number of days with trades (buys or sells), which ends up assigning a higher weight to funds with higher turnover.

Figure 1. Post-trade returns relative to counterfactual

This figure presents differences between average factor-neutral returns of stocks bought/sold and those of random buy/sell counterfactual strategies for buy and sell trades. The bracket at the top of each bar is the 95% confidence interval of the point estimate at each horizon. Confidence intervals in brackets are computed using double-clustered standard errors, calculated as described in Appendix A.4.



month, quarter, or year. We regress these measures on averages of our trade-based buy and sell performance measures over the same horizons.²⁸ Importantly, both the buying and selling performance measures are strongly linked with benchmark-adjusted performance. This is consistent with poor selling imposing nontrivial opportunity costs on the PM's potential performance. Given that PMs in our sample have fairly long holding periods, coefficients on our performance measures increase as the horizon increases; the coefficients on both measures range between 0.22-0.25 over a 1 year horizon. In the right panel, we see that results are unchanged if we control for the average difference between the hold portfolio and the benchmark, which is somewhat similar to an overall attribution measure that can be computed with a single snapshot of holdings.

²⁸Similar results obtain if we weight portfolios by the length of time for which we observe data or use value-weighted performance measures in place of our baseline.

Table 2. Linking buying and selling-based performance measures with fund benchmark-adjusted performance

This table presents the association between factor-adjusted portfolio returns in excess of the benchmark and our measures of buying and selling performance. A fund's average performance at holding, buying, and selling (by our measures) is highly associated with the magnitude by which they beat their benchmark in the cross-section, even after adjusting for exposures to value, size, market, and momentum factors. Buy, sell, and hold portfolios are adjusted for factor exposures to value, momentum, size, and market, calculated for the forward-looking horizons, the Buy - Hold, Hold - Sell, and Hold - Benchmarks are computed for each fund date and then averaged across the full sample by fund. Fund returns are similarly calculated on a rolling basis and then averaged over the full period. Fund returns are not adjusted for factor exposures, and neither are benchmarks. Benchmarks are assigned based on what the fund manager has chosen for use with the Inalytics service.

	Dependent variable: Average benchmark-adjusted portfolio return over					
	1 Month (1)	1 Quarter (2)	1 Year (3)	1 Month (4)	1 Quarter (5)	1 Year (6)
Hold - Benchmark				0.123*** (0.031)	0.131*** (0.029)	0.110*** (0.026)
Buy - Hold	0.061** (0.022)	0.120* (0.049)	0.250*** (0.074)	0.057** (0.021)	0.100* (0.046)	0.245*** (0.069)
Hold - Sell	0.094*** (0.021)	0.086** (0.029)	0.227** (0.072)	0.082*** (0.021)	0.079** (0.029)	0.239*** (0.072)
Observations	750	750	749	750	750	749
R ²	0.053	0.041	0.073	0.105	0.096	0.107

Note:

*p<0.05; **p<0.01; ***p<0.001

Intriguingly, estimated coefficients on our trade-based performance measures are larger than the Hold-Benchmark coefficient at longer horizons.

As with any performance evaluation exercise, a number of questions emerge about the robustness and interpretation of our main results. Accordingly, we have conducted a battery of additional tests to address several of potential concerns. Table 3 reports results from a subset of these tests in which we consider a variety of alternative ways of constructing our performance measures, either by restricting the set of holdings/trades used in the analysis or by changing the way in which we construct the relevant long-short portfolios. See Appendix A.5 for a more detailed discussion and additional tests.

First, we illustrate the implications of altering the approach for selecting securi-

Table 3. Post-trade returns relative to counterfactual, overall and robustness checks

This table presents the average value added measures (post-trade returns relative to a random sell counterfactual) for buy and sell trades for a variety of specifications. See text and Appendix A.5 for additional details.

Performance Measure	Panel A: Buy					Panel B: Sell				
	28	90	180	270	365	28	90	180	270	365
(1) Factor-Neutral	0.34 (0.04)	0.58 (0.09)	0.83 (0.15)	0.98 (0.20)	1.16 (0.24)	0.03 (0.04)	-0.21 (0.09)	-0.50 (0.15)	-0.69 (0.18)	-0.80 (0.20)
(2) Value-weighted*	0.30 (0.05)	0.63 (0.11)	0.99 (0.23)	1.23 (0.32)	1.45 (0.39)	-0.02 (0.04)	-0.31 (0.10)	-0.71 (0.21)	-0.92 (0.26)	-1.04 (0.29)
(3) Omit recently added stocks from hold portfolio*	0.35 (0.04)	0.63 (0.09)	1.94 (0.15)	1.14 (0.21)	1.39 (0.27)	0.01 (0.04)	-0.26 (0.09)	-0.61 (0.14)	-0.78 (0.17)	-0.91 (0.20)
(4) Unadjusted Return	0.39 (0.05)	0.71 (0.10)	0.92 (0.18)	1.08 (0.22)	1.27 (0.27)	0.06 (0.04)	-0.12 (0.10)	-0.42 (0.17)	-0.59 (0.20)	-0.70 (0.23)
(5) DGTW Adjusted	0.30 (0.04)	0.45 (0.09)	0.43 (0.15)	0.48 (0.19)	0.55 (0.24)	0.04 (0.04)	-0.07 (0.09)	-0.31 (0.13)	-0.48 (0.17)	-0.49 (0.20)
<i>Match counterfactual by:</i>										
(6) Benchmark-adjusted return quintile*	0.32 (0.05)	0.57 (0.10)	0.85 (0.16)	1.09 (0.22)	1.32 (0.28)	0.03 (0.05)	-0.19 (0.09)	-0.45 (0.12)	-0.66 (0.17)	-0.79 (0.20)
(7) Idiosyncratic volatility quintile*	0.35 (0.05)	0.59 (0.08)	0.87 (0.15)	1.02 (0.20)	1.10 (0.24)	-0.01 (0.05)	-0.29 (0.09)	-0.61 (0.15)	-0.86 (0.18)	-0.98 (0.21)
(8) Position size quintile*	0.26 (0.04)	0.41 (0.09)	0.56 (0.13)	0.72 (0.19)	0.93 (0.25)	0.05 (0.04)	-0.21 (0.09)	-0.52 (0.13)	-0.71 (0.16)	-0.86 (0.19)
(9) Omit 5% Smallest*	0.19 (0.04)	0.36 (0.09)	0.35 (0.14)	0.46 (0.19)	0.46 (0.24)	-0.08 (0.04)	-0.19 (0.10)	-0.69 (0.14)	-0.71 (0.18)	-0.86 (0.21)
(10) Omit 5% Most Illiquid*	0.35 (0.05)	0.61 (0.10)	0.88 (0.15)	1.06 (0.20)	1.28 (0.25)	0.02 (0.05)	-0.25 (0.10)	-0.50 (0.15)	-0.57 (0.20)	-0.73 (0.23)
(11) Omit returns for 7 days post-trade*	0.18 (0.03)	0.40 (0.09)	0.67 (0.15)	0.81 (0.19)	0.91 (0.24)	-0.05 (0.04)	-0.31 (0.09)	-0.61 (0.15)	-0.80 (0.19)	-0.90 (0.22)

* Indicates that results are factor-neutral (where not otherwise specified).

ties for our no-skill random selling counterfactual. The first approach in row 2 uses weights proportional to lagged portfolio values in place of equal weights for the hold portfolio—analogue to selecting a security at random with a probability proportional to size or selling all holdings pro-rata—and weights multiple buy/sell trades on the same day proportionally to transaction size. The second approach in row 3 uses additional available information to construct the counterfactual. It is rare for PMs to sell stocks that were recently added to the portfolio (see Appendix Table A.8). Thus, we construct a counterfactual hold portfolio which excludes stocks which are in the bottom quintile of the distribution of holding length at each date. Both adjustments yield modest increases in the buying performance measure and modest decreases in the selling performance measure.

Second, a natural concern is that stocks traded tend to have above average exposures to systematic factors linked with expected returns relative to stocks held, meaning that our estimates could be driven by risk compensation rather than skill. Indeed, any performance evaluation exercise involves a joint hypothesis about whether the benchmark is correct. While our long-short construction differences out many systematic exposures which are common between stocks traded and held, there is always a possibility that we are sorting on characteristics which are already known to drive expected returns, presumably because they proxy for priced systematic factors. Specification 4 of Table 3 also reports estimated return measures from repeating the analysis in Figure 1 using raw cumulative returns. Results are quite similar between our baseline and this unadjusted specification, so the explicit role of the Carhart risk adjustment procedure on our overall estimates is small or nonexistent. We also leverage financial information from the Worldscope database to construct characteristics following the approach of Daniel et al. (1997) (row 5), matching counterfactual trades based on joint quintiles by size, book to market, and momentum characteristics. Rows 6-7 match counterfactuals by quintiles of characteristics which are potentially correlated with propensity to sell assets (as we show below)—prior returns over the previous quarter and idiosyncratic volatility.²⁹ While each exercise changes magnitudes to some modest extent, results

²⁹Further, to allow for the possibility that PMs adjust position sizes partly to reflect ex-ante expectations about benchmark-adjusted returns from systematic factors, we also demonstrate that results are quite similar if we select the counterfactuals within the same quintile by position size (row 8). We discuss the role of conviction further in section 4.5 below. See also Appendix A.5 for many additional tests. Appendix A.9 shows that we find similar results if we express long-short returns

are quite consistent across all specifications: both the annualized outperformance of buys and underperformance of sells are always statistically significant and economically meaningful in magnitude.

Third, we consider the potential role of unobserved transaction costs in driving our results. Perhaps the hold portfolio includes many illiquid stocks which would be too costly (whether due to direct trading costs or indirect costs from price impact) to sell. In this case, our estimates of value-added relative to a feasible counterfactual potentially neglect any differences in these transaction costs between the security traded and our proposed alternative. To address this concern, we have conducted a battery of tests which drop the smallest (by market capitalization) or least liquid (by Amihud liquidity measures) stocks in our sample (see e.g., rows 9-10) and/or form counterfactuals matched within-portfolio by quintiles of these measures. Similar results obtain if we omit the smallest or least liquid stocks from counterfactual portfolios only.

In addition, it is possible that PMs' actual buy and sell trades could be sufficiently large so as to generate price impacts. In that case, we could measure a change in the price of the security actually traded but fail to incorporate a corresponding change which would have been induced in the counterfactual portfolio if it was traded instead. Mean reversion in prices would tend to bias both performance measures downward. Our approach to addressing this concern is simple: since these direct price impacts are largely transitory, we can compute performance starting several days after the transaction takes place, which allows for transitory effects to dissipate (row 11).³⁰ Our results are not meaningfully affected across these specifications.³¹

in the modal currency of each portfolio rather than USD. In addition, Appendix A.10 conducts a complementary analysis in which we form long-short calendar time portfolios on the basis of signals which are constructed using our trading data. These portfolios, which *buy* stocks recently sold by PMs and short sell stocks recently held, earn very high Sharpe ratios due to substantial benefits of diversifying across (largely uncorrelated) positions traded by different PMs. For several reasons which we discuss further there, we prefer our baseline approach to the calendar time portfolio method. However, these results are consistent with our main analysis, and these portfolios earn Sharpe ratios which are consistent with the degree of diversification implied by standard errors on our baseline point estimates.

³⁰Further note that the timing over which underperformance of selling, which is most pronounced at longer horizons, manifests is inconsistent with a simple price impact explanation.

³¹While these analyses help to rule out a number of concerns related to transaction costs and potential price impacts, we acknowledge some potential limitations of these tests. First, we do not observe intraday data, so our analysis cannot capture profits which were captured at a very high frequency (see, e.g., Puckett and Yan 2011). As a result, while the daily frequency of our data does allow us to rule out favorable intraquarter trading as driving our results, we are unable to directly test

Finally, Appendix A.5 also presents several empirical tests which address the potential role for position size limits and tracking error budgets in driving our main results. Once again, we find little evidence consistent with PMs facing constraints which “force” sales which underperform a random selection of existing holdings. Perhaps most convincingly, we find almost no correlation between the “Buy - Hold” portfolio and the “Sell - Hold” portfolio (between 1.5% and 2.5%) for buys and sells on the same days, suggesting that there are not systematic factor exposures matched between buys and sells within portfolios. If a PM were trying to maximize return within a tracking error constraint, we would expect trades on each side to have common factor exposures, so as to not accidentally tilt the fund away from its target benchmark.

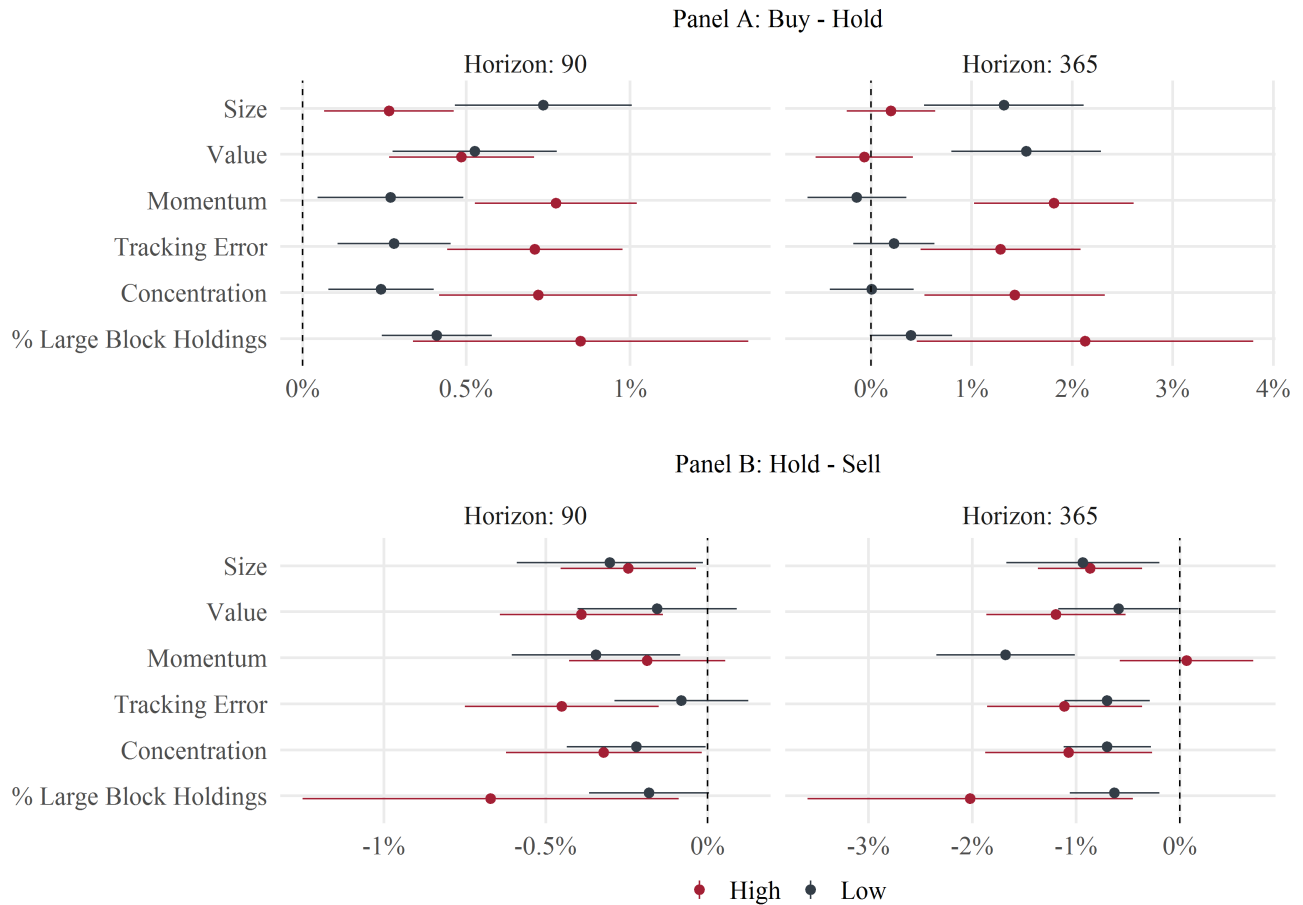
3.3 Heterogeneity

Thus far we have presented results based on averages across all PMs. In this section we consider potential sources of heterogeneity in average buying and selling performance across PMs in our sample. Figure 2 presents performance results based on a number of covariates such as turnover, tracking error, and investment style. We find that underperformance in selling appears most prominently amongst fundamentals-oriented managers who hold more active, highly concentrated portfolios with higher tracking error. PMs in this space may have fewer procedures in place to systematically process information about existing holdings (e.g., one might expect these PMs to prioritize soft information relative to quantitative relative valuation signals). In other words, if buys are less related to systematic information, it might be less natural to have effective systems to assist with selling decisions.

whether sales are executed at particularly favorable intraday transaction prices. While this could be the case, we do observe that both our buying and selling measures predict benchmark-adjusted portfolio performance across short and longer horizons. Moreover, if unobserved trading costs were driving our results, then basic comparative statics logic would imply that matching on observable liquidity proxies and timing adjustments should significantly dampen the effect. We do not find evidence for this prediction. In this sense, our analysis parallels an approach taken in the literature which tries to test for adverse selection in insurance markets (see, e.g., Einav and Finkelstein 2011, for a survey) by testing for selection on observables. Thus, price impacts of the average individual trade that are large in magnitude and persistent over substantial periods of time (e.g. longer than 1 week) could potentially change the interpretation of our selling results. Given the size of the portfolios (which is still fairly small relative to total institutional trading volume), the average trade size, and the results which omit illiquid securities and microcaps, we consider this possibility unlikely, but it is difficult to rule out completely.

Figure 2. Variation in Buying and Selling Skill by Manager Characteristics

The end of the bar is the 95% confidence interval of the point estimate at each horizon. Confidence intervals are computed using double-clustered standard errors, calculated as described in Appendix A.4. Details on characteristics construction can be found in Appendices A.2 and A.5.



With respect to style, funds that score higher on momentum appear to perform better in selling; funds that score higher on value appear to underperform most in selling. Whereas selling is a key component of the momentum strategy, value plays—at least in terms of how they are described heuristically—are often more about identifying undervalued assets, which seems psychologically more closely aligned with buying. For further details, see Appendix A.2.

4 Explaining Underperformance

In this section, we propose a potential mechanism linking the use of heuristics to systematic underperformance of selling strategies relative to a feasible counterfactual. We then provide evidence for the mechanism by exploiting the panel nature of our database to ask whether patterns in funds' actual trading strategies are associated with predictable differences in performance.

We argue that our results can be explained by expertise with the asymmetric allocation of cognitive resources: PMs focus more on buying and are prone to 'fast', heuristic decisions when selling. This conjecture implies that 1) shifts in cognitive resources towards the latter has the potential to improve performance and 2) heuristics associated with limited attention are more likely to be observed for selling than buying.

4.1 Potential Reasons for an Asymmetric Allocation of Attention

Our interviews with PMs suggest that decisions are overwhelmingly focused on the buying domain relative to the selling domain. Why might this be the case? Several institutional features offer partial explanations. Perhaps the most obvious reason is that, as noted above, virtually all managers in our sample are fundamental rather than quantitatively oriented and have very long holding periods. Research, especially in-depth research that involves constructing custom valuation models and qualitative judgements via conversations with management of various companies, is costly and difficult. The fact that search processes emphasize large blocks of unstructured interactions between PMs and clients (see, e.g., [Goyal et al. 2020](#)) suggests that being able to present new investment ideas in a compelling narrative is an important qualitative signal which helps PMs to win and keep existing business. As such, PMs may devote the lion's share of their cognitive resources towards finding the next winner to add to the portfolio.

At the same time, a manager who is buying based on private relative valuation judgements can also sell based on similar valuations. In an "information-ratio" centric world, where managers are evaluated based on bets relative to a benchmark, it is straightforward to map short-selling type positions to an equivalent underweighted position relative to a benchmark. For owned positions that make up a small propor-

tion of the benchmark, relative value judgements still inform which stocks are best to eliminate from a portfolio.

In turn, even taking as given that the client-manager relationship may lead to an initial asymmetric allocation of attention, the question remains of why professional PMs have not learned that their selling decisions are underperforming simple no-skill strategies. Indeed, Table 2 suggests that poor selling has a substantial drag on benchmark-adjusted returns, one of the key metrics which affects PMs' ability to attract clients (Del Guercio and Tkac 2002). The environment in which fund managers make decisions offers several clues. As Hogarth (2001) notes, learning requires frequent and consistent feedback. While it is feasible to generate this type of feedback for both buy and sell decisions, Appendix A.6 discusses how common reporting standards are much better suited to identify underperformance in buying decisions than selling decisions. Thus, PMs and their clients are more likely to receive frequent, valid feedback about their purchases than their sales, which can explain the failure to learn about underperformance in the latter domain.

Importantly, a PM may have two potential reasons to sell a stock: first, she could receive negative information about the stock and subsequently decide that its current price is not justified, and second she could need to raise cash to finance a new buy. Our results point to the second type of selling decision as being worse relative to a counterfactual strategy of simply trimming equal amounts from every other position (on either a value-weighted or equal-weighted basis). The first type of trade seems to perform well; as we discuss in section 4.3, sales on earnings announcement days, when PMs are likely to be paying more attention to their trades, perform significantly better than on other days.

4.2 Bounded Rationality in Selling

How might bounded rationality affect PMs' trading decisions? Many models of decision-making in psychology (Hauser and Wernerfelt 1990) and economics (Lleras, Masatlioglu, Nakajima, and Ozbay 2017) split choices between multiple alternatives—in our case, choosing what asset(s) to buy and sell—into two stages: generating a consideration set and then selecting an option from that set. Prior work has shown that cognitive constraints can lead to the use of heuristics in both stages of the process (Hauser 2014).

For example, [Barber and Odean \(2008\)](#) posit a two-stage process for both buying and selling decisions, where limited attention constrains the consideration set to assets with salient attributes and biases in preferences lead to potentially suboptimal choices from that consideration set.³² We outline how a two-stage decision-model would operate in our setting and provide evidence for bounded rationality in both stages of the selling process but, importantly, not the buying process. We then demonstrate how the specific heuristics we document can potentially explain the results presented in the preceding sections, including the underperformance of sales relative to a random-sell counterfactual.

In the first stage, rather than considering the entire portfolio, we posit that limited attention places bounds on the consideration set ([Hirshleifer and Teoh 2003](#)). Research in psychology and economics has found that these consideration sets are often determined by the ranking and filtering of objects based on salient attributes.³³ Information on prior returns is ubiquitous, and according to models of salience ([Bordalo, Gennaioli, and Shleifer 2013](#)), the high variation around average returns should make this attribute particularly top-of-mind for a fast-thinking PM.³⁴ In turn, extreme deviations in relative returns in either the positive or negative direction naturally emerge as candidates for filtering assets to be included in the consideration set.

In the second stage, the PM must choose which assets from the consideration set to trade. According to the attribute substitution framework of ([Kahneman and Frederick 2002](#)), people making ‘fast,’ heuristic decisions may replace the more difficult question of “which asset in this set is least likely to outperform in the future” with an easier question to answer, such as “how much conviction do I have in this position?” or “how well do I understand this company?” This attribute substitution process can potentially generate systematic underperformance if it leads PMs to, for example, sell still-viable investment ideas.

³²Also see [Sakaguchi, Stewart, and Walasek \(2017\)](#) for how a two-stage model explains the disposition effect.

³³See [Lleras et al. \(2017\)](#) for an overview of such attention-based filtering in decision-making. For example, in consumer choice [Gourville and Soman \(2007\)](#) find that people faced with options that differ along several attributes end up only considering those that rank on the extreme ends of those dimensions.

³⁴Consistent with this, former investment banker and Bloomberg columnist Matt Levine writes “The rule of thumb wisdom for buying is about fundamentals, but for selling it’s usually about price action.” <https://www.bloomberg.com/opinion/articles/2019-01-10/investors-have-to-sell-stocks-too>.

Consistent with attention-based filtering in the first stage, we first show that exogenous events which potentially *expand* PMs' consideration sets by drawing attention to current holdings—earnings announcement days—are associated with substantially better selling performance. We then demonstrate that positions with extreme returns are indeed over-represented in PMs' consideration sets: assets that are in the top or bottom 5 percent based on prior returns are nearly 50 percent more likely to be sold relative to those that just over- or underperformed. Importantly, consistent with the conjecture that more attention is channeled towards buying than selling, earnings announcement days are not associated with changes in buying performance and prior returns have no detectable association with purchase decisions.³⁵ Finally, we show that low conviction assets are significantly more likely to be sold from the PMs' consideration sets and that the systematic sale of these assets can potentially explain the associated underperformance.

Note that, at first glance, these results might appear to contrast with those of [Barber and Odean \(2008\)](#), who find that retail traders' buying decisions are more prone to attention-based heuristics than their selling decisions. We argue that the two sets of results are complementary once one considers important differences in the groups' decision environments and levels of expertise. [Barber and Odean \(2008\)](#) argue that salient cues are more likely to affect buying decisions because it is much more difficult to attend to the entire set of potential buying opportunities than the limited number of stocks in one's portfolio. On the buying side, PMs are less likely to display attention-driven heuristics because of the process involved in purchasing an asset. As is described in Section 2.1, managers spend a great deal of time and resources cultivating an investment universe that includes the set of stocks that they are considering at any given time. Active research identifies potential mispricing and investment opportunities, at which point the PM will add the respective assets to their portfolio.³⁶ Moreover, institutional investors are likely to possess expertise that retail traders do not; as a result, when PMs' cognitive resources are devoted to a decision, those deci-

³⁵The results from [Hartzmark \(2014\)](#) offer additional support for our proposed mechanism of expertise with asymmetric allocation of cognitive resources: retail investors, who tend to be less sophisticated overall, appear to make both their selling *and* buying decisions based on extreme returns.

³⁶This process is described at length in David Swensen's bestselling text titled "Pioneering Portfolio Management: An Unconventional Approach to Institutional Investment" ([Swensen 2009](#)), and is consistent with interviews of PMs in our sample.

sions are considerably more likely to outperform those of retail investors.³⁷ Thus PMs' purchases are less likely to be heuristic-driven because of the combination of expertise and resources devoted to the decision. On the other hand, if PMs are devoting few resources to their selling decisions, those choices are more likely to be driven by attentional heuristics. Notably, while the retail traders in Barber and Odean (2008) hold 4.3 assets on average, PMs in our sample hold 78.5. While it is possible even for an attention-constrained individual to consider the entire portfolio when it only includes 4.3 stocks, this is less likely to be the case for a portfolio that is almost 20 times larger. In turn, we would expect that the consequence of asymmetric resource allocation towards buying decisions would manifest as attention-based heuristics on the selling side.

4.3 Performance on announcement days

We gather earnings announcement dates from the I/B/E/S database and recompute our counterfactual return strategies for stocks which are bought/sold on those days, relative to all other trading days.³⁸ Managers have a strong incentive to pay close attention to stocks in their portfolios on these dates for several reasons. As discussed in Section 1, the information in financial statements, associated press releases, and conference calls (which even offer opportunities for managers to directly address questions to the company) provide a wealth of new pieces of hard and soft information that are decision-relevant and can potentially improve trading performance (Easley et al. 2008). This information is both (relatively) easily available and salient, since earnings announcement dates are known in advance, and results are heavily covered by the financial press. In turn, we conjecture that earnings announcements prompt PMs to broaden their consideration sets of what to sell, potentially mitigating the constraints imposed by attentional heuristics.

Panel II of Table 4 presents the difference in average performance of trades on announcement versus non-announcement days. Panel A reports the difference be-

³⁷This is reflected in the stark difference in overall performance. While PMs in our sample largely outperform their respective benchmarks, retail traders actually underperform the market (Odean 1999).

³⁸Our results do not change if we look at performance of trades within a 1, 2, 3, or 4 day window of the announcement.

Table 4. Average Trading Performance Differential: Earnings vs. Other Days

This table presents the the difference between averages of our value added measures for trades of stocks on their earnings announcement days versus all other days (I), and we report the difference between average performance of buys and sells for trades on announcement dates vs non-announcement dates using the baseline measure (II). Double-clustered standard errors, computed using the method described in Appendix A.4, are reported in parentheses.

Panel A: Buy						Panel B: Sell				
Horizon	28	90	180	270	365	28	90	180	270	365
I. Average difference earnings vs. other days:										
Factor-neutral	-0.04	0.14	0.24	0.70	-0.32	0.37	0.29	1.08	2.05	1.54
	(0.21)	(0.39)	(0.62)	(0.81)	(1.02)	(0.20)	(0.37)	(0.58)	(0.77)	(0.80)
Unadjusted	-0.14	0.08	0.16	0.62	-0.25	0.28	0.40	1.16	2.13	1.60
	(0.21)	(0.40)	(0.65)	(0.85)	(0.98)	(0.19)	(0.39)	(0.62)	(0.80)	(0.95)
II. Average performance difference of buys and sells:										
Non-announcement trades						0.31	0.79	1.33	1.68	1.97
Announcement trades						-0.12	0.56	0.41	0.20	0.05

tween average value-added of buy trades executed on earnings announcement days compared with average value-added from all other buy trades.³⁹ There is little systematic difference in performance, and whatever differences exist are not statistically significant. This is consistent with attentional resources already being devoted towards purchase decisions; information released on earnings announcement days is carefully incorporated into purchase decisions just like other forms of information is used on non-announcement days. Panel B demonstrates the stark contrast in the performance of selling decisions on announcement versus non-announcement days. Selling decisions on announcement days add substantially more value than those sold on non-announcement days. Our point estimate of this difference is 154 basis points over a one year horizon in the baseline specification, with all estimates at the 180 day horizon and onwards being significant at the 5% level. These results hold for raw returns as well.

³⁹Given the much smaller number of observations associated with stocks sold on earnings announcement dates, the average performance of sells on earnings announcement dates is positive but imprecisely estimated. Accordingly, we emphasize and report differences between average returns on non announcement days rather than levels. Point estimates for non-announcement days are virtually identical to the overall numbers in section I of Table 3. When computing a standard error for the difference between the two estimates, we impose the assumption that the covariance between the two estimates is zero. This is likely conservative, given that most likely the two estimates are positively correlated (e.g., because stocks sold on earnings announcement days might also be sold several days later as well), which would have the effect of reducing the standard error on the difference.

Finally, Panel II of Table 4 compares the difference between average performance of buys and sells on earnings announcement days versus non-announcement days. In the baseline specification, our point estimate of selling performance on announcement days is +79 bp at a 1 year horizon, which is only 5 bp lower than the corresponding estimate for buying performance on those days. In contrast, buys outperform sells by nearly 200 bp at a 1 year horizon on all other dates, and differences in relative magnitudes are similarly large at other horizons as well.

These findings suggest that when contemporaneous predetermined events shift PMs' attention towards existing positions—leading them consider a wider set of assets and information than they otherwise would when deciding what to sell—their selling performance improves substantially. In Appendix A.8, we present complementary evidence consistent with better selling performance when PMs are more likely to attend to these trades. In particular, we argue that larger transactions are more likely to be 'slow' decisions that are attended to. Indeed, we show that the largest quintile of sales based on transaction size does not underperform the counterfactual at longer horizons, while all other quintiles underperform. Together, this evidence suggests that the overall poor selling performance is does not seem to be due to a fundamental lack of skill in selling.

4.4 Predicting Buying and Selling Decisions

We now attempt to provide more direct evidence that attention constrains PMs' consideration sets in what to sell—but not in what to buy—by looking at the relationship between trade decisions and prior returns.

4.4.1 Measuring association between prior returns and trade decisions

For each portfolio-date, we identify a set of stocks (a subset of holdings in the prior day's portfolio) potentially under consideration to be bought or sold, rank existing holdings according to past benchmark-adjusted returns, and then ask whether managers are more likely to trade based on these ranks. Given the size of our dataset, we adopt a flexible, non-parametric approach to measuring managers' tendency to buy and sell positions based on past returns. Specifically, for the set of prior holdings which are included in the analysis, we compute a measure of returns, usually relative to the benchmark over the same horizon. For each portfolio and trading date, we sort prior

positions into 20 bins using these relative rankings, where we set the breakpoint between bins 10 and 11 equal to zero, so that all stocks in bin 10 and below have declined relative to the benchmark. We also emphasize within-manager rankings, rather than absolute levels of these measures, since the definition of “extreme returns” depends on the types of assets in a given PM’s investment opportunity set.⁴⁰

While this approach is straightforward for a long-only PM’s selling decisions given that the consideration set for sell trades is composed of the current holdings, constructing the consideration set for buying decisions requires taking additional factors into account. Our first approach looks at purchases of assets that already exist in the portfolio; this captures the majority of buys, including large ones (adding up to 99 percent to existing holdings). Our second approach, described in Appendix A.7, includes all purchase decisions—including the opening of brand new positions—and calculates relative prior returns by broadening the consideration set to assets that are likely being considered for purchase.

For this exercise, our preferred measure of prior returns is computed as follows. For positions which were opened more than 1 quarter (90 days) prior to the date of interest, we use the benchmark-adjusted return of the stock from 90 calendar days prior through the trading day before the date of interest. For positions with shorter holding periods, we change the starting point for computing the benchmark-adjusted return to the opening date.⁴¹ We use this as our preferred measure because performance is often reported to clients at a quarterly frequency, and, from a more pragmatic perspective, this construction is less sensitive to fact that position opening dates are left-censored. However, as we show in Section 4.4.2, results are robust to alternative definitions of past returns.⁴²

⁴⁰For details on constructing the bins, see Appendix A.7

⁴¹Results are robust to a wide variety of horizons and whether of not gains and losses are computed since purchase or over a fixed horizon. For example, in Appendix Table A.7, we report probabilities of buying using a prior return measure which does not depend on the time of initial purchase and does not impose the restriction on holding length. In that specification, our main results on the relationship between average buying probabilities and prior returns maintain.

⁴²We find nearly identical results if we restrict attention to stocks with opening dates that are observed during our sample. To avoid a fairly mechanical relationship between our prior return measure and the probability that a manager will add to/reduce an existing position from splitting trades across adjacent days (see Appendix A.7), we exclude stocks that were bought in the very recent past. Related to this concern, our regression analyses below always control for the holding period since the position was opened and the holding period since last buy, as well as squared terms of each.

4.4.2 Buying and selling based on past returns

We present results as fractions of positions that are bought or sold within each of the prior return bins. These fractions, which can be interpreted as probabilities, are computed by first calculating the proportion of stocks sold within each bin at the fund-date level, then averaging across all fund-dates in the sample. Figure 3 depicts the results for selling and buying decisions of assets that are already held graphically using a variety of different prior return measures, with 20 bins formed on each measure. Bins are sorted from left to right according to prior returns. We begin with the buying probabilities. The probability of purchasing a stock already held is quite flat across the bins of prior returns. These results hold across all prior return measures considered and no pronounced patterns appear as we move towards more extreme bins in all cases.

A very different picture emerges for the selling probabilities. Assets with more extreme relative returns are substantially more likely to be sold relative to stocks in the central bins. An asset with a prior return in one of the most extreme bins is more than *50 percent* more likely to be sold than an asset with a less extreme return. Moreover, assets in these most extreme bins (1 and 20) have much higher selling probabilities than adjacent bins; such discrete jumps are altogether absent for buying probabilities. Despite the fact different specifications use prior return measures calculated over a variety of horizons, a very pronounced U-shape appears across all.

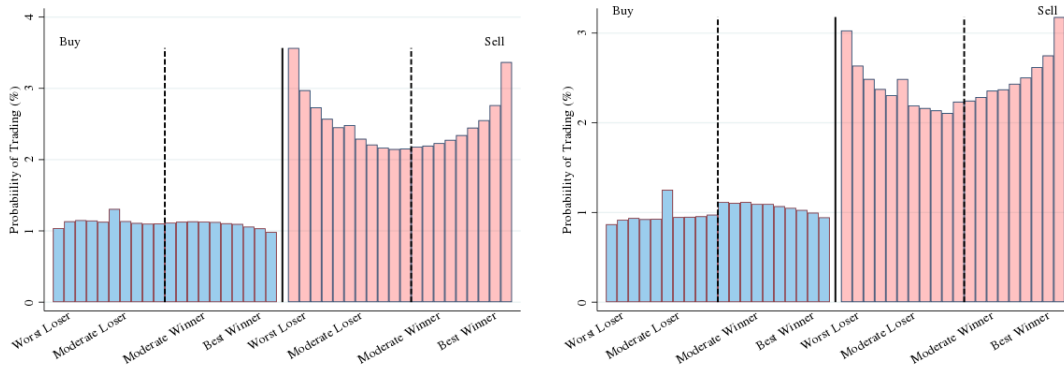
In Appendix A.7 and Appendix Table A.7, we report two additional results on the relationship between prior returns and buying/selling strategies. First, we extend the analysis to allow for a wider consideration set for purchases; this enables us to incorporate additions of new stocks into the analysis. As above, we find essentially no relationship between prior returns and the likelihood that a stock is purchased. In addition, we look at the relationship between making large purchases or sales (which more than double or halve the existing portfolio weight) and prior returns. Again, we find a strong relationship between extremeness of prior returns and selling probabilities, while seeing no such relationship for purchases.

The U-shaped selling pattern is similar to the rank effect in Hartzmark (2014), where retail traders are more likely to buy *and* sell assets with extreme returns. Hartzmark (2014) documents a similar rank effect in mutual funds as well. However, due to

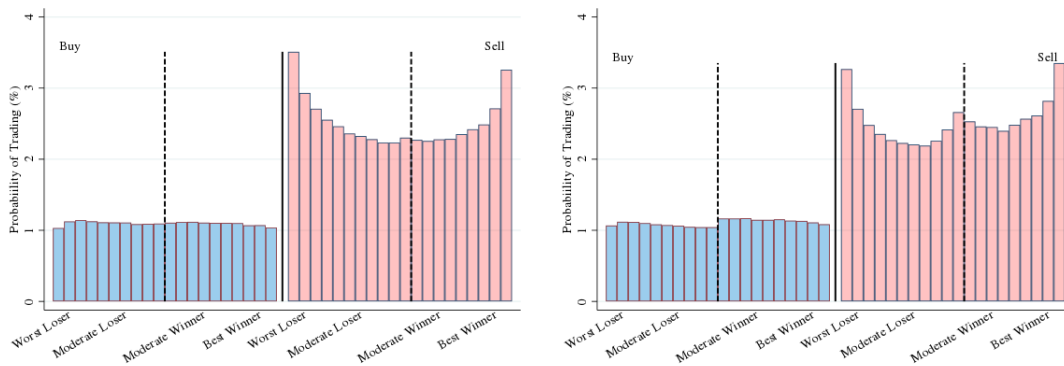
Figure 3. Probability of buying and selling based on past returns

This set of figures reports daily buying (blue) and selling (red) probabilities for stocks in the portfolio sorted into 20 bins by various past return measures. Panel A sorts on cumulative past benchmark-adjusted returns since the purchase date or one quarter/year, whichever is shortest. Panel B sorts on past benchmark-adjusted returns of a position over one quarter and one year. Panel C sorts on past raw returns of a position over one week and one day. The ten bins on the left are positions with negative returns and the ten bins on the right are positions with positive returns.

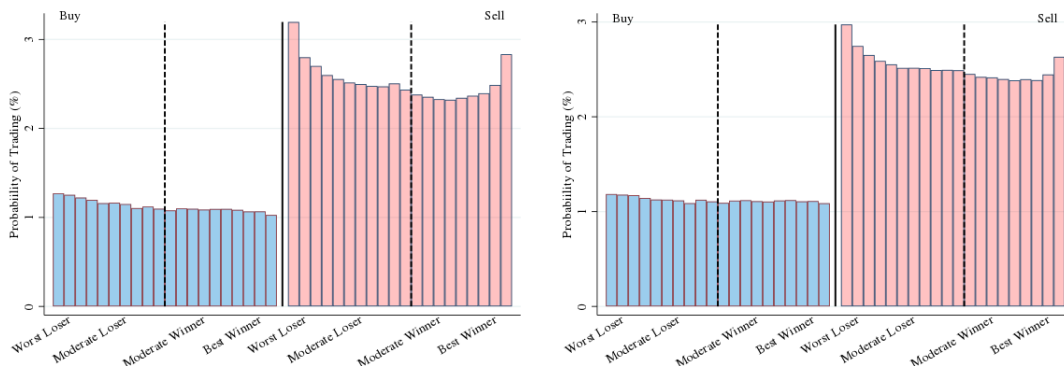
Panel A: Cumulative benchmark-adjusted returns capped at 1-quarter and 1-year



Panel B: Past benchmark-adjusted 1-quarter and 1-year returns of a position



Panel C: Short-horizon 1-week and 1-day returns



the use of quarterly data, he notes that the behavior can be driven by strategic concerns in response to investor preferences. We show that in contrast to these findings, expert PMs display this heuristic when selling but not when buying. Additionally, as outlined in Section 5, our ability to see each manager's entire portfolio and trades over time allows us to study whether this heuristic is costly for institutional investors. The richness of the dataset also permits exploring the underlying psychological mechanism, yielding evidence that the heuristic is driven by an asymmetric allocation of attention and that its costs are not driven by the unloading of extreme positions *per se*, but rather by the types of positions sold from the constrained consideration set.⁴³

4.4.3 Alternative explanations

We now consider several instrumental reasons that could potentially explain our results. As discussed in Section 2, the vast majority of portfolios in our sample are tax-exempt, so the U-shaped selling pattern cannot be rationalized with tax concerns. Our finding that sales are more likely for positions with extreme returns over very long (1 year) and very short (1 week) horizons makes agency-based explanations—where PMs are reluctant to report realized losses to their clients—unlikely. Agency-based explanations also seem unlikely to explain the large jumps in probabilities observed between the 19th and 20th (1st and 2nd) bins relative to the 18th and 19th (2nd and 3rd) bins. These jumps are consistent with limited attention constraining the consideration set of what to sell, as the top and bottom 5 percent of returns are much more likely to be displayed and made salient to PMs (see [Ungeheuer \(2017\)](#) for direct evidence). This also mitigates concerns about risk management motives, since the relative risk of assets in extreme bins is likely to be fairly comparable to less extreme adjacent bins.⁴⁴

We also examine whether our observed pattern can be explained by other variables

⁴³[Ben-David and Hirshleifer \(2012\)](#) document a related phenomenon where retail investors are more likely to sell securities with high or low prior returns; [An \(2015\)](#) shows that stocks with such returns are more likely to outperform in the following month. This V-shaped trading pattern is distinct from ours in that the high or low returns are calculated at the level of the individual security rather than with respect to overall portfolio returns. Moreover, the pattern is documented both for buying and selling decisions.

⁴⁴In subsequent regression analyses, we will include controls for idiosyncratic volatility, systematic factor exposures, and position size, all of which are potentially relevant for risk management. Inclusion of these controls generally has a very limited impact on estimates analogous to the nonparametric statistics presented above.

which could be correlated with our prior return measures: holding length and position size. We find that both the U-shaped selling pattern and the flat buying pattern is observed across the holding length and position size variable sorts. For more detailed results, see Appendix A.7.

Finally, Table 5 reports estimates from a series of linear probability models for the likelihood of selling or buying, which allow us to control for a number of time-varying fund characteristics (either via controls, fund fixed effects, or fund-date fixed effects), calendar time effects, as well as other position characteristics. All specifications include linear and quadratic controls for holding length since the position was opened, holding length since last buy, and position-level portfolio weight (as a fraction of total portfolio assets under management). The key regressors of interest are dummies for each of the prior return categories, which have the same interpretation as the bins used in the preceding analyses, where the omitted category remains bin 10 (slight loser positions). Results are similar with different prior return measures and different numbers of bins.

We begin with the right panel which characterizes selling probabilities. Across all of these specifications, the difference in the predicted probability of selling a stock in bin 20 is at least 50 percent higher than the probability of selling a stock in bins 6 through 15, and always considerably higher than bin 19. Likewise, we observe similar strong nonlinearities for stocks in bins 1 through 2 relative to more central bins. Column 4 includes stock-date fixed effects as in Hartzmark (2014), so the main coefficients of interest are identified via variation in the relative return categories across portfolio managers who hold the same stock on the same date. Even then, we find that positions in the most extreme returns bins are substantially more likely to be sold.

Turning to the left panel, the relationship between buying probabilities and prior return measures is much more muted. In the loss domain, most of the coefficients are insignificant despite being estimated on a sample of over 50 million observations. Even the significant coefficients are substantially smaller in magnitude than the coefficients associated with selling probabilities. In the saturated specification presented in column 5, only the coefficient on bin 1 is statistically distinguishable from zero. Looking at large positive returns, we observe a greater number of significant coefficients, but the differences between central and extreme bins (e.g., bins 16 through 18 and bin 20 or bins 19 and 20) are very small relative to the same differences for sales.

Table 5. Probability of trading based on prior returns

This table presents position-level estimates of a linear probability model (in percentage points) for the likelihood of trading a given stock. The key explanatory variables of interest are indicators corresponding to 20 bins of past benchmark-adjusted returns capped at one year, where the tenth bin is the omitted category. We control for fund characteristics including lagged log(AUM), prior-month turnover, the annual volatility of a funds benchmark-adjusted returns, and prior month loadings on Fama-French Cahart regressions (calculated using the Dimson (1979) procedure using 1 year of prior daily returns). We adjust for position-level characteristics including linear and quadratic terms in holding lengths (overall and since last buy) and position sizes(% AUM) at the beginning of the day. The coefficients and t-statistics are reported for the estimates on each bin. We present versions of these estimates which condition on trading (buys for buying probability, sells for selling probability) in Table A.9 in Appendix A.7.

	Buying Probability				Selling Probability			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bin 1	-0.041 (-1.789)	-0.047* (-2.068)	-0.074** (-3.178)	-0.144* (-2.192)	1.389*** (14.828)	1.379*** (14.592)	1.237*** (13.032)	0.796*** (6.343)
Bin 2	0.040* (2.071)	0.033 (1.685)	0.008 (0.410)	-0.058 (-1.217)	0.806*** (12.580)	0.799*** (12.330)	0.666*** (10.471)	0.543*** (6.212)
Bin 3 to 5	0.046** (3.275)	0.041** (2.817)	0.025 (1.606)	0.001 (0.031)	0.432*** (10.998)	0.428*** (10.731)	0.331*** (8.980)	0.308*** (5.686)
Bin 6 to 9	0.025** (2.856)	0.022* (2.337)	0.011 (1.164)	0.012 (0.763)	0.109*** (6.750)	0.107*** (6.328)	0.067*** (4.549)	0.079** (3.262)
Bin 11 to 15	-0.029** (-2.701)	-0.031** (-2.679)	-0.032*** (-3.568)	-0.014 (-0.771)	0.028 (1.560)	0.037* (1.966)	0.005 (0.250)	0.105*** (4.213)
Bin 16 to 18	-0.088*** (-4.699)	-0.092*** (-4.668)	-0.109*** (-6.291)	-0.152*** (-3.679)	0.312*** (8.004)	0.318*** (7.945)	0.234*** (6.063)	0.559*** (8.760)
Bin 19	-0.129*** (-5.810)	-0.133*** (-5.769)	-0.155*** (-7.276)	-0.220*** (-4.071)	0.578*** (10.849)	0.582*** (10.725)	0.485*** (9.125)	0.834*** (9.505)
Bin 20	-0.169*** (-6.700)	-0.175*** (-6.665)	-0.212*** (-8.366)	-0.286*** (-4.179)	1.186*** (15.869)	1.186*** (15.638)	1.071*** (14.303)	1.132*** (9.581)
Fund Control	Yes	Yes	No	Yes	Yes	Yes	No	Yes
Fixed Effects	None	Date	Fund × Date	Stock × Date	None	Date	Fund × Date	Stock × Date
R ²	0.022	0.028	0.283	0.281	0.005	0.009	0.179	0.317
N	54.2M	54.2M	56.2M	45.5M	54.2M	54.2M	56.2M	45.5M

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Taking stock, the regression specifications, in conjunction with the nonparametric evidence in Appendix Table A.8, suggest that the considered sources of omitted variable bias are unlikely to explain our results. Additionally, Table A.9 calculates the same model, conditional on a trade (at least one buy on a given day for a given fund when analyzing buys, and at least one sell when analyzing sells). We find that the results are qualitatively similar, but with much larger magnitudes for the selling decisions. Together these results are consistent with non-instrumental motives driving selling but not buying decisions.

4.5 Selling and Conviction

We now examine which assets from the consideration set PMs choose to sell. As discussed above, the attribute substitution framework of [Kahneman and Frederick \(2002\)](#) predicts that people making heuristic decisions may replace the difficult question of determining which asset is least likely to outperform with an easier question related to conviction or psychological attachment. To capture this, we order positions based on their *active share*, or how much they are overweighted relative to the benchmark ([Cremers and Petajisto 2009](#)). This measure captures how much the PM stands to gain if the stock beats the benchmark.⁴⁵ Assets with high active share typically correspond to positions that the manager has spent a good deal of effort building up over time, becoming familiar and attached to the firm in the process. This costly procedure likely generates greater conviction in the position for non-instrumental reasons, such as sunk costs or psychological ownership ([Anagol, Balasubramaniam, and Ramadorai 2018](#); [Heath 1995](#); [Kahneman, Knetsch, and Thaler 1990](#)).

Positions with low active share can manifest for three main reasons: 1) a position had a high active share but has done very poorly, 2) the PM has added a position to the portfolio but has not built it into a larger one, or 3) the PM chooses to hold a position close to the benchmark weight of a stock which is large in the benchmark (or underweight, though negative active shares are fairly uncommon).⁴⁶ The PM may

⁴⁵ Active share is calculated by taking the difference between a PM's weight on a stock in the fund and subtracting the corresponding weight, if any, of the same stock in the client-provided benchmark, a measure which is provided to us by Inalytics. Since performance is evaluated based on benchmark-adjusted returns, an asset that is overweighted generates excess returns when it goes up and excess losses when it goes down.

⁴⁶ A fourth alternative is that the PM has actively reduced a formerly large position through prior

Table 6. Probability of selling by active share and past returns

This table reports differences in probabilities, in percentage points, of selling by bins of past benchmark-adjusted returns double sorted with bins of position-level active share, relative to a baseline category (the tenth bin of past benchmark-adjusted returns, within each active share quartile). Columns represent different active share bins, along with the difference across rows between the smallest active share bin and the average across the other bins (active share bins 2-4). Calculations for 8 categories of prior returns, formed from 20 bins of past position returns, are reported in rows. We report the baseline selling probability for the 10th bin in the last row.

Prior Return Bins	Active share Bins				Lowest - Others
	Lowest	Low	Higher	Highest	
1	2.264	0.724	0.452	0.414	1.734
2	1.501	0.445	0.320	0.223	1.171
3-5	0.828	0.280	0.150	0.105	0.650
6-9	0.254	0.122	0.051	0.013	0.192
11-15	-0.067	0.093	0.115	0.146	-0.185
16-18	0.159	0.387	0.424	0.488	-0.274
19	0.580	0.689	0.790	0.859	-0.199
20	1.426	1.338	1.360	1.410	0.057
Baseline Level: Bin 10	3.329	1.851	1.691	1.779	1.556

still have conviction in a stock in the first category as she had exerted time and effort in building it up in the past. Rather than reflecting a particular view about future returns, assets in the third category may be in place to minimize exposure of a fund's relative returns to the idiosyncratic returns of large assets in the benchmark.

In contrast, assets in the second category are most likely to include the PM's 'neglected ideas.' The manager has gathered enough favorable information on each asset to add a position to the portfolio, but has not elected to build it up further. In turn, heuristic thinking would generate fewer reasons to keep a low active share asset in this category. Moreover, time-constrained PMs may monitor them less closely than higher conviction positions, especially if other factors draw attention elsewhere (e.g. attractive buying opportunities). Thus, in discarding these positions, PMs may be throwing out still-viable investment ideas—ones potentially capable of outperforming many existing holdings—especially if high conviction positions are more likely to have already realized their anticipated upside potential.

sells.

Table 6 documents the PMs' propensity to sell based on active share, both overall and within each of the 20 bins of prior returns. To construct this table, assets within each portfolio are sorted into four bins based on their active share. We then construct a measure capturing the propensity to sell an asset based on its prior returns; specifically, the difference in the probability of selling a stock in a given bin of prior returns relative to the middle one (bin 10, Slight Loser). The last column reports the difference between the lowest active share bin and the average across the other three active share bins in the same row of prior returns. We report baseline probabilities for the omitted bin in the last row.

Results are consistent with PMs being most prone to selling neglected ideas, as measured by low active share, from the consideration set of extreme returns.⁴⁷ First, examining the baseline probabilities, we note that low active share positions are substantially more likely to be sold regardless of the level of prior returns. Second, we find that stocks in the lowest active share bin are much more likely to be sold when they exhibit prior returns below the benchmark, especially extreme ones, relative to high active share assets. The probability of selling a stock with the lowest active share and lowest prior return bin is 5.6, or 140 percent larger than the baseline probability of selling, which is 2.3. Assets in these bins are also 155 percent more likely to be sold than those which experienced similar levels of underperformance (bin 1) but have the highest active share. Thus, low active share positions are particularly likely to be discarded when they are in the consideration set of extreme underperformance. Selling probabilities in the lowest active share bin are relatively less affected by moderate gains, and responses to the most extreme gains in bin 20 are similar regardless of active share.

We then examine whether sales of low active share assets tend to underperform relative to a random selling counterfactual. Panel A of Table 7 depicts the performance of sales relative to a random counterfactual by bins of stocks' active share for our baseline value-added measure. As above, we sort positions into four bins based on a prior day estimate of active share, then separately form counterfactual performance measures for these different subsamples of trades. We see a stark contrast in performance: assets in the lowest three active share bins underperform substantially more

⁴⁷Consistent with results in the prior section, we find that buying probabilities do not exhibit a significant relationship with prior returns.

Table 7. Post-trade sell returns relative to counterfactual by active share

This table presents the average factor-neutral returns relative to random sell counterfactuals for sell portfolios sorted by active share. We compute average returns of stock held minus returns of stocks sold. We rank the active share measures within funds at a daily level and sort them into four bins from Lowest, Low-Med, Med-High to Highest sizes. For the Lowest Active Share bin, we further split into two halves by a position's weight: Smaller positions (below the 50th percentile) and Larger positions (above the 50th percentile), after sorting by active share. Columns represent sell performance measures at the following horizons: 1 month, 3 months, 6 months, 9 months, and 1 year. We report point estimates of average counterfactual returns for each portfolio at different horizons. Heteroskedasticity and autocorrelation robust standard errors, computed using the method described in the section A.4, are reported in parentheses.

Active Share	Horizon				
Bins	28	90	180	270	365
Lowest, Smaller Positions	-0.46 (0.10)	-0.96 (0.32)	-1.46 (0.66)	-1.80 (0.91)	-1.62 (0.96)
Lowest, Larger Positions	0.10 (0.07)	0.01 (0.13)	-0.05 (0.21)	0.14 (0.29)	0.55 (0.33)
Low-Med	0.20 (0.07)	-0.15 (0.18)	-0.47 (0.35)	-0.86 (0.41)	-1.19 (0.58)
Med-High	0.16 (0.07)	0.13 (0.13)	-0.15 (0.19)	-0.24 (0.26)	-0.84 (0.34)
Highest	0.32 (0.08)	0.29 (0.21)	0.23 (0.39)	0.13 (0.50)	-0.35 (0.56)

than sales of assets in the top active share bin, where the latter performs much closer to the the counterfactual. This is consistent with PMs holding on to high active share assets when thinking fast; thus, when sales of those positions are observed, they are more likely to be informed ones.

As discussed above, very low active share positions may be held to hedge a fund's exposure to idiosyncratic returns of large stocks in the benchmark. These positions would appear in the data as having a very low active share and a high portfolio weight.⁴⁸ In turn, we separate stocks in the lowest active share bin into two categories based on absolute position size, and recompute our counterfactual performance measures for these subsamples. Whereas large positions with low active share tend to have positive

⁴⁸As an example, if Apple is 3% of a PM's benchmark index we might observe a position in Apple of 3% which would have an active share of zero, despite the fact that 3% would be a quite large absolute position size.

counterfactual performance measures (consistent with these being somewhat passive positions and good candidates for sales), we find that sales of assets from the second ‘neglected ideas’ category—low active share and low position size—to substantially underperform.

It is tempting to conclude from these results that since the underperformance of selling strategies is associated with low active share (usually smaller) positions, the costs in terms of overall portfolio performance associated with these transactions is likely to be small. However, this reasoning is incorrect provided that changes in portfolio weights induced by selling smaller initial positions are similar to those from selling larger ones. Holding trade size as a fraction of portfolio market value constant, the cost in foregone profits from a suboptimal trade are independent of the initial size of the position.⁴⁹ Indeed, we find that average trade sizes for sells are quite similar across both active share and initial position size bins. Further, recall from Table 2 above that our measures of average selling performance are highly predictive of overall benchmark-adjusted performance.

In addition, we note that poor selling is likely associated with performance in two ways. The first is a direct effect: it changes the weights in the current portfolio. The second is an indirect effect: ‘fast’ selling of low conviction positions may lead those stocks to be discarded from the consideration set of future buys. Consistent with the potential importance of this latter effect, we find that, once sold, assets are substantially less likely to be purchased again, suggesting that attention-constrained elimination of small positions may interfere with a PM’s security selection process.⁵⁰

Note that it is possible that our results may be explained by an alternative mechanism which can generate underperformance of sales as a function of only the first stage outlined above. Specifically, if extreme positions are ranked similarly across a large enough number of institutional portfolios and sold for reasons unrelated to funda-

⁴⁹Further, since the effect of an idiosyncratic stock return on overall portfolio variance is a convex function of the weight, one could argue that the effect on measures of performance that adjust for idiosyncratic risk exposures such as the information ratio are larger for small positions.

⁵⁰Before a security has been sold once, there is a 76% chance that the portfolio will purchase a security at least once more. After the first time a portfolio sells a security, there is only a 40% chance the portfolio will ever increase its position in that security again before liquidating it. Once a stock is removed from the portfolio, there is only a 40% chance that the position is ever added back to the portfolio. On the intensive margin, the daily probability that a stock is purchased again declines from 3.3% to 2.4% once a stock is sold for the first time.

mental value, then the increased propensity to sell them can generate downward price pressure on current prices. Although the decision to unload an asset by an individual investor may have been driven by contextual factors specific to their portfolios, if the asset is categorized as extreme in enough portfolios, then this would lead to performance being a function of prior returns at the market level. [An \(2015\)](#) and [An and Argyle \(2016\)](#) provide evidence from holdings of individuals and mutual funds suggesting that, as selling pressure is reduced over the medium term, subsequent mean reversion in prices can lead these stocks to systematically outperform on a risk-adjusted basis going forward.⁵¹

5 Heuristic use and Performance

In this section, we exploit the panel nature of our dataset in order to illustrate a more direct link between the performance of selling strategies and fund-level characteristics, such as the propensity to sell assets with extreme returns. As in [Section 3](#), we compare the returns of the actual stocks traded with counterfactual random selling strategies. We then ask whether patterns in funds' actual trading strategies are associated with predictable differences in performance. To operationalize this, we compute several fund-level characteristics and sort fund-weeks into categories based on these characteristics, then compute the average value-added associated with PMs' trades in each bin. Before proceeding, we note that this analysis is only able to identify correlations in the data, so it is not feasible to rule out other time-varying fund characteristics which simultaneously drive performance and observable properties of trading behavior.

We begin by considering the impact of heuristic use on performance. Based on the mechanism outlined in [Section 4.1](#), we use the greater propensity to sell assets with extreme returns as a proxy for heuristic use.⁵² To capture what we term 'heuristic intensity,' positions are sorted into 4 bins based on the fraction of stocks sold that are located in the extreme bins (Worst Loser and Best Winner) for each fund-week.⁵³

⁵¹At the same time, as we discuss further in [Appendix A.4](#), we find very little correlation between stocks sold at the same time in our dataset. These correlations are slightly positive for stocks sold on the same day and slightly negative at other leads and lags, which provides some evidence against this channel in our data.

⁵²This greater propensity is only a proxy for heuristic use because, as demonstrated in [Section 4.1](#), PMs do not randomly sell assets from the consideration set of extreme returns.

⁵³For instance, the mean of this heuristics intensity measure is around 0.4 on a monthly basis, which

In order to reduce noise in the measures related to discreteness, we only compute the heuristic intensity measure for funds which hold at least 24 stocks (i.e., there are 4 stocks on average per bin) and only include fund-weeks in which at least 3 stocks are traded.⁵⁴ We then rank fund-weeks into four categories according to this measure to calculate relative performance of the associated selling decisions. Our primary rationale for a weekly frequency is that it provides a satisfactory balance between reducing potential noise in the sorting variable (by averaging over multiple trades) while still operating at a high enough frequency to capture variation in heuristic-use.

Panel A of Table 8 presents sample averages of counterfactual returns sorted into four bins based on heuristic intensity within-fund. Over the course of each PM's time series, each week is sorted into one of four categories based on its level of heuristic intensity. We find that our proxy for heuristic intensity is positively associated with significant underperformance. The highest levels of heuristic intensity are associated with the worst performance, especially at the longer horizons. Magnitudes are quite substantial: at a 1 year frequency, the highest level of heuristic intensity predicts an average of around 220 foregone basis points relative to a random-sell counterfactual. At the same time, average performance of sales which occur when PMs are selling fewer extreme positions is statistically indistinguishable from the counterfactual. Appendix A.8 demonstrates the robustness of our results when using the unadjusted performance measures. In contrast to these results, we find no similar relationship when looking at differences in average heuristic use *between*-managers, a result which further supports our assertion that underperformance in selling, rather than being driven by persistent differences in skill, is due to the asymmetric allocation of attention.⁵⁵

In the preceding section we argued that both stages of the selling process are prone to heuristic thinking—limiting the consideration set to assets with salient attributes and then choosing to sell those that the PM has least conviction in. The literature on heuristics and biases documents that people are more likely to rely on heuristics during

would imply (through a simple application of Bayes' rule) that the likelihood of a stock being sold in the extreme bin is 4/3 the likelihood of a stock being sold in one of the central bins. In Appendix Table A.12, we show, perhaps surprisingly, our measure of heuristics intensity is largely uncorrelated with a variety of observable fund characteristics.

⁵⁴We lose about one quarter of fund-week observations due to these restrictions.

⁵⁵More specifically, during periods of time when the PM is attending to sales and less reliant on heuristics, the performance of her selling decisions increases substantially.

Table 8. Post-trade sell returns relative to counterfactual by fund behavior

This table presents average returns relative to random sell counterfactuals for sell portfolios sorted by heuristics intensity, cumulative benchmark-adjusted fund returns since the beginning of a quarter, and a proxy for ‘cash raising’ episodes. We divide these measures into four bins from Lowest, Low-Med, Med-High and Highest, based on their rankings. Columns represent sell performance measures at the following horizons: 1 month, 3 months, 6 months, 9 months, and 1 year. Double-clustered standard errors, computed using the method described in the section A.4, are reported in parentheses.

Fund Characteristics	Bins	Horizon				
		28	90	180	270	365
Panel A: Heuristics Intensity Fraction of extreme stocks sold weekly (sorted across funds)	Lowest	-0.07 (0.07)	-0.21 (0.14)	-0.29 (0.20)	-0.42 (0.24)	-0.28 (0.33)
	Low-Med	0.00 (0.06)	-0.07 (0.11)	-0.16 (0.15)	-0.12 (0.21)	0.03 (0.24)
	Med-High	0.05 (0.06)	0.05 (0.14)	-0.13 (0.21)	-0.20 (0.22)	-0.31 (0.26)
	Highest	0.11 (0.09)	-0.59 (0.19)	-1.33 (0.31)	-1.57 (0.39)	-2.20 (0.47)
Panel B: Cumulative Benchmark-adjusted Fund Return since the beginning of a quarter (sorted across funds)	Lowest	-0.22 (0.10)	-0.96 (0.24)	-1.43 (0.38)	-1.81 (0.48)	-1.95 (0.56)
	Low-Med	0.01 (0.06)	-0.07 (0.13)	-0.46 (0.19)	-0.50 (0.28)	-0.69 (0.30)
	Med-High	-0.02 (0.07)	-0.15 (0.12)	-0.42 (0.21)	-0.69 (0.26)	-0.59 (0.31)
	Highest	0.16 (0.09)	0.12 (0.17)	0.21 (0.29)	0.03 (0.40)	0.06 (0.43)
Panel C: Net Buy Weekly number of stocks bought minus number of stocks sold (sorted within fund)	Lowest	-0.01 (0.08)	-0.47 (0.18)	-0.73 (0.26)	-1.25 (0.36)	-1.44 (0.46)
	Low-Med	0.04 (0.08)	-0.20 (0.17)	-0.73 (0.23)	-0.97 (0.36)	-1.15 (0.40)
	Med-High	0.04 (0.07)	0.11 (0.13)	-0.01 (0.19)	-0.08 (0.24)	-0.35 (0.29)
	Highest	0.08 (0.06)	-0.03 (0.13)	-0.26 (0.20)	0.01 (0.25)	0.23 (0.28)
Panel D: Flows Weekly Assets entering portfolio as % of AUM (sorted within fund)	Most outflows	-0.06 (0.09)	-0.53 (0.18)	-0.86 (0.30)	-1.10 (0.38)	-0.99 (0.46)
	Low-Med	0.05 (0.06)	-0.12 (0.12)	-0.47 (0.18)	-0.56 (0.24)	-0.73 (0.29)
	High-Med	0.08 (0.06)	-0.12 (0.12)	-0.27 (0.18)	-0.53 (0.23)	-0.79 (0.27)
	Most inflows	0.07 (0.07)	-0.09 (0.13)	-0.42 (0.20)	-0.54 (0.28)	-0.71 (0.33)

situations when cognitive resources are in higher demand, such as in times of stress or when attention is otherwise occupied (see [Kahneman \(2003\)](#) for review). Panels B and C of Table 8 consider two empirical proxies intended to capture periods emblematic of such episodes. As in Panel A, these measures are computed on a weekly basis and sort fund-weeks into four categories to capture either between or within-manager variation.

The first aims to capture performance when the PM is likely to be stressed. Institutional investors are known to take stock of their own performance based on calendar time, e.g. on a quarterly or yearly basis. Based on the conjecture that the PMs are more likely to be stressed when their overall portfolio is underperforming, we construct a measure that captures portfolio performance relative to the beginning of the preceding quarter. Table 8, Panel B demonstrates that selling quality is worst (relative to a random-sell counterfactual) when the PM's overall portfolio is underperforming the most. Panel C considers a measure that proxies for sales that are likely to be driven by cash raising considerations rather than forecasts of relevant performance metrics. We posit that observing larger bundles of assets being sold (relative to being bought) is emblematic of the manager being in “cash-raising mode.” We compute the difference between the number of stocks bought and the number of stocks sold, where both measures are expressed as fractions of the number of stocks in the portfolio. We find that the difference between the number of stocks bought and sold predicts greater underperformance of the selling decisions.⁵⁶

As discussed above, some institutions such as mutual funds face direct pressure from performance-driven fund flows, creating the possibility that flows induce PMs to engage in large sales under unfavorable conditions ([Alexander et al. 2007](#)). Following [Coval and Stafford \(2007\)](#), we construct daily flows as the portion of assets under management which are not explained by the previous day's returns, then aggregate them to a weekly frequency. We test the above hypothesis in Panel D by creating proxies for flows and sorting fund-weeks into bins based on these proxies. Consistent with our discussion in Section 2.1 that flow concerns are less pronounced in our institutional

⁵⁶In similar spirit, we present additional evidence in Appendix A.8 which relates performance to turnover. There, we sort each portfolio's time series into periods with low and high turnover. We find that performance of sales deteriorates during high turnover periods (which are presumably characterized by better trading opportunities), while performance of buys remains unchanged. This is consistent with a mechanism where better buying opportunities attract the already scarce attentional resources allocated to sales, further degrading performance in that domain.

context, the performance results—that buys outperform the counterfactual while sells underperform—are unchanged across the flow measures.

6 Conclusion

We utilize a unique dataset and find evidence that financial market experts—institutional investors managing portfolios averaging \$573 million—display costly, systematic heuristics. A striking finding emerges: While investors display skill in buying, their selling decisions underperform substantially—even relative to random-sell strategies. We provide evidence that investors use heuristics when selling but not when buying, and that these heuristic strategies are empirically linked to the documented difference in performance.

As shown in Section 3, the comparison of trades on earnings announcement versus non-announcement days suggests that PMs do not lack fundamental skills in selling; rather, results are consistent with PMs devoting more cognitive resources to buying than selling. When decision relevant information is salient and readily available—as it is on announcement days—PMs’ selling performance improves substantially. We propose a mechanism through which overall underperformance in selling can be explained by a heuristic two-stage selling process, where PMs limit their consideration set to assets with salient characteristics (extreme prior returns) and sell those they have the least conviction in. A proxy for this heuristic strategy is associated with substantial losses relative to a no-skill random selling strategy.

In light of the imbalance in feedback discussed in Section 4.1 and Appendix A.6, the theoretical framework of Gagnon-Bartsch, Rabin, and Schwartzstein (2018) suggests that PMs may fail to recognize their underperformance in selling even in the long-run. The authors show that a mistaken theory such as the favorability of selling positions with extreme returns may persist in the long run because people channel their attention through the lens of this theory. As in Schwartzstein (2014), errors persist due to the person ignoring information that seems irrelevant and only updating her beliefs based on information that is attended to. Our findings imply significant benefits to creating environments where learning can occur more effectively, such as through amended reporting standards that emphasize counterfactual sales performance. Moreover, our empirical results on a link between heuristic use and underperformance of

selling strategies suggest that PMs adoption of decision aids and/or simple alternative selling strategies may substantially improve performance.

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A Appendices

A.1 Data construction: additional details

The primary source of our analysis is Inalytics’ holdings data and changes in holdings. We apply two primary filters to select the set of portfolios to include in our analysis. First, we drop portfolios for which daily trading data are unavailable or appear to be incomplete.⁵⁷ Second, we exclude funds that do not have a sufficient fraction (at least 80 percent) of portfolio holdings which could be reliably matched with CRSP or Datastream. In demonstrating the robustness of our results, we perform the analyses using data for larger stocks (using market cap information from Worldscope) and those with higher trading volume. These markets arguably have better price discovery and higher match rates with CRSP/Datastream.

After we clean the holdings data, we convert all the prices into USD using exchange rates from Datastream. To ensure accuracy in exchange rates, we compare the exchange rate in Datastream with two other sources of exchange rates from Compustat and

⁵⁷Trades are sometimes imputed at month-end because Inalytics receives portfolio snapshots in adjacent months which do not fully match the aggregated the trade data. This requires a reconciliation process. Because of this, we exclude funds that have a large fraction of trades occurring at the end of each month.

Inalytics. In the event of a discrepancy, we pick the two out of three that are the same, and this procedure takes care of discrepancy in all cases. We then augment the holdings data by merging in external prices series and returns from CRSP (US stocks), Datastream (International stocks) and Inalytics' provided price series in this order. The external price series allow us to compute the market value of each holding precisely. There are instances where the market value of a stock (likely due to a measurement error in price/quantity) seems implausibly high, so we employ an iterative weight cleaning algorithm to eliminate these positions from the analysis. We provide additional details about these steps below.

We begin by outlining the key steps of our data cleaning procedure:

1. **Cleaning identifiers:** Inalytics has four main types of identifiers for stocks: SEDOL, ISIN CUSIP, and LOCAL. For the first three types of identifiers, they are distinguishable by the number of digits. SEDOL has 6-7 digits, CUSIP has 8-9 digits, and ISIN has 12 digits. In a few instances one type of identifier is mislabeled by the clients, so we correct them according to the number of digits.
2. **Merging in liquidated stocks with holdings data:** There are instances when a fund completely closes a position, so a stock disappears from the holdings data. Since our main trade measure is computed from the change in stock's holding, a position-closing trade will not be observed in the holdings. To do so, we first measure the minimum and maximum dates of each fund. Then, we locate the instances where the stocks disappears from the portfolio between the start and end dates. We then append those stocks back to the holdings data.
3. **Dropping portfolios without daily trades:** Some of the portfolios in the dataset do not receive daily time-stamped trade data. In these cases, only monthly holdings are reported and trades are imputed at the end of the month. To filter out these portfolios, we count the fraction of trades after the 27th of the month for each fund. If a fraction of trades after the 27th for a fund is over 50% or missing (in case of no trades observed), we drop the portfolio from the analysis sample. In addition, Inalytics independently provided a list of these portfolios from their internal records, essentially all of which were filtered out by this criterion. We also remove these manually flagged portfolios.

Next, we discuss some potential issues related to measurement errors in the price data. Where available, we use external price series from CRSP and Datastream, and we

supplement the remainder with data provided by Inalytics. Inalytics relies on multiple data vendors, including MSCI and Thompson Reuters, as well as clients themselves, for price series in the holdings data. In some cases, reported prices are overstated, which would lead us to incorrectly characterize portfolio weights and potentially introduce measurement errors in various counterfactual return calculations. We rely on our external price series as the primary measure for a price when computing returns and portfolio weights throughout the analysis, only relying on Inalytics where we otherwise would lack a security price. Our performance measures only include stocks which have a valid price series for the full holding period.

When we compute cumulative returns for purposes of evaluating trading performance, we winsorize extremely small and large return realizations, some of which may be due to measurement errors in the price data. To mitigate the effect of the extreme returns when computing the average returns, we winsorize returns in the holdings dataset across all measures (raw and factor neutral) before forming portfolios. In our baseline results that we present here, we employ two winsorizing thresholds. First, we winsorize the cumulative return measures on each date across all positions at 0.1% on either tail. As an additional precaution, we winsorize large positive returns in the whole sample at the 99.99% threshold on the right tail of the distribution for raw returns and 0.01% on either tail for factor neutral returns. The rationale is that factor neutral returns can have also have extreme negative returns after adjusting for risk, so it is necessary to winsorize on both tails for risk-adjusted returns measures. We have also considered larger thresholds for winsorizing such as 0.3%, 0.5%, and 1% and obtain similar results.

In a handful of cases (e.g., because a stock split has led to an incorrectly high market value), the market value of a single position appears to be extremely large relative to the rest of the portfolio, which is indicative of a likely measurement issue. In order to flag situations when one errant price could cause our estimates of portfolio weights to be substantially biased, we employ an iterative procedure to drop potentially problematic positions. In essence, we look for situations where the entire portfolio is concentrated in a single, extremely large position. For these purposes, we compute the market value of a position as the minimum of raw Inalytics price and raw external prices times the quantity of stocks, and then compute the position-level weight by dividing through by the dollar value of all positions. With these weights in hand, the procedure proceeds as follows. First, we compute the first three largest weights at a portfolio-date level.

We then compute two measures 1) the difference between weights of the largest and second largest-held stocks and 2) the difference between weights of the second and third largest-held stocks. If the first difference minus the second difference is over 15%, the largest weight is over 10% and the second difference is less than 5%, we flag the stock with the largest weight, and exclude it from the analysis and the weight calculation. We then recompute stocks' weights after the largest-held stocks are dropped and repeat the procedure to flag other stocks with unusually high weight in a portfolio. We repeat this algorithm until there is no stock with an unusually large weight in portfolios. This iterative weight-dropping algorithm finishes in 5 runs. There are 57,982 stock-date observations to be excluded from weight calculation. 84.3M observations (94.12%) in the holdings data have no weight errors. The first run of this algorithm cleans up weights for 4.1M observations (4.62%) in the holdings data. After five runs of this algorithm, whereby we exclude five stocks at most, 99.86% of holding observations have no weight problems. There are two portfolios for which this procedure still indicates the presence of a handful of extremely concentrated positions but these two portfolios only made up 39,128 out of 89M observations.

We use the Reuters Worldscope database of equity characteristics for information other than daily prices and volume. When we need to perform filters using market cap, compute characteristics quintiles for DGTW-style adjustments, or perform any other analysis based on the fundamentals (e.g., book-to-market ratio) of the securities, we use this merged sample, which matches 80% of our full sample. When we perform for liquidity adjustments, we merge in daily volume from Datastream, which has data from global exchanges converted to USD. Datastream volume covers 83% of our daily holdings data, and we use this subsample when constructing liquidity-based subsamples or dropping securities below a certain Amihud measure threshold.

The Amihud measure is calculated on a monthly basis using daily dollar volume (in USD) from Datastream and returns from our sample of security returns (described earlier in the appendix). We use the prior month's worth of daily data to construct the illiquidity measure, which is the average of the absolute value of daily returns over that period divided by the dollar volume. Because we are interested in keeping the most liquid stocks in the counterfactual, we use the set of securities for which this measure is available in our matched sample when calculating a 5% breakpoint.

In Table 2, where we analyze whether our buying and selling skill measures are associated with the excess return of the fund as a whole, we compute the Buy - Hold

and Hold - Sell portfolios as we do in the rest of the paper, and subsequently take averages of those measures over all dates. Hold - Benchmark is the value of the hold portfolio return (over some future window) less the benchmark return over the equivalent window. The buy, sell, and hold portfolios are adjusted for factor exposures to value, momentum, size, and market. Fund returns are similarly calculated on a rolling basis and then averaged over the full period. Fund returns are not adjusted for factor exposures, and neither are benchmarks. Benchmarks are self-designated by PMs/clients when initiating the Inalytics monitoring service. Inalytics provides daily benchmark-adjusted return series, which we compound over the relevant period for purposes of the regression.

We construct a measure of net flows in and out of a given strategy at various horizons. The flow is estimated as the remainder of total assets under management not explained by the prior returns of the fund. In other words, if we had 100 dollars in a fund, we observe a period return of 20%, and the subsequent period had 140 dollars in the fund, we consider the \$20 remainder as the net flow of additional AUM injected over the period. Note that changes in net cash positions will look like flows according to our measure, since we do not observe cash positions in the holdings data.

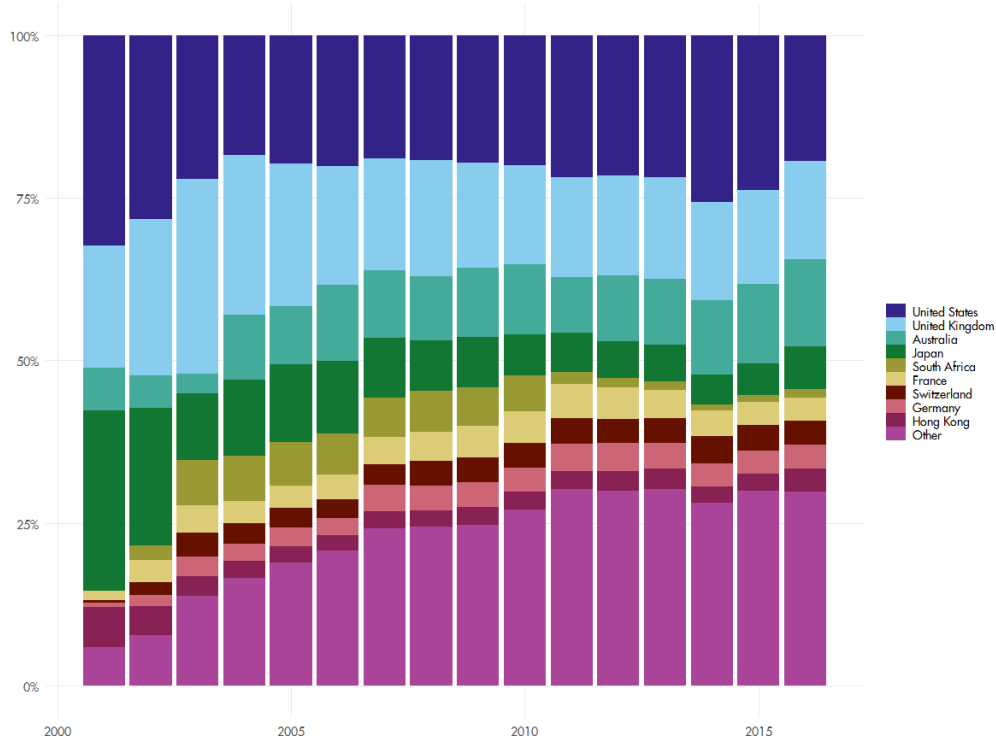
A.2 Additional descriptive statistics

In order to better understand the sample, we collect information on the portfolios by hand. Because we only observe names of the portfolios as they are entered into Inalytics' system (not, for example, the specific traceable vehicles that are readily merged with external data), we are only able to identify a subset of manager and portfolio names. Often these portfolios are associated with specific separately managed accounts which are constructed with custom characteristics for clients, and thus may not have a related public strategy. Portfolios can have names which lack obvious identifying characteristics, making them impossible to match to a specific firm or strategy.

Where possible (70% of the sample), we identify the management company, and hand-classify them as Fundamental, Quantitative, Other, or some combination based on the way that they describe themselves on their website. We find only 22 managers in the sample are purely quantitative, with the remainder being primarily fundamental (sometimes with quantitative filters performed before an analyst reviews a universe). The vast majority describe their investment process as being built around custom re-

Figure A.1. Breakdown of portfolio holdings by country over time

This figure presents a holdings-level breakdown of the sample over time. Much of the variation in the sample is driven by the fact that firms enter and exit the data over time, and so (for example) the Japan-focused managers are diluted as more firms enter the sample. The sample is based on the Worldscope classification of a security's domicile, and aggregated at the fund level as a percent of daily AUM, before being averaged by year across managers.



search from internal analysts, with specific recommendations derived from proprietary analysis focused on individual companies. Managers consistently self-describe as having high-conviction, which is also verifiable on the basis that they only hold around 2/3 of the number of stocks as the average US-based mutual fund. This implies a set of highly active portfolio managers, selected for specific markets by institutional clients, who are meant to hold concentrated portfolios that deviate substantially from their benchmarks.

Unlike the managers studies in the majority of the mutual fund literature, our sample is not only focused on US equities. Many managers focus on their own domestic markets, including mandates focused on Europe, UK-specific, Australia, South Africa, and Middle Eastern markets, in addition to the US. Figure A.1 presents a breakdown of the average portfolio geographic exposure by year.

Our portfolios are also different from other databases of separately managed accounts used in the prior literature. For example, the managers in our sample have lower annualized portfolio turnover—while the median annualized portfolio turnover is 35% in the Inalytics sample, it is 59% in the Morningstar Principia database (Chen et al. 2017). Managers in our sample also tend to be more concentrated, holding fewer names at a time. The average manager in the Inalytics database holds 78 distinct stocks, versus 103 stocks for managers in the Principia database (Chen et al. 2017). The portfolios in our sample also have lower average loadings on traditional style factors like value, size, and momentum, perhaps reflecting that these factors may be country specific rather than global in nature (Griffin 2002), and that our sample contains mandates which focus on specific regions other than the United States.

Next, we discuss some of the summary statistics which we compute from our position-level data and report in Panel B of Table A.1. Our simplest position-level variable is an indicator variable which equals 1 if the manager buys or sells a given stock on a given date. Of the 89 million position-date combinations in our sample where a stock was in the portfolio at either the start or end of the day, about 2.4 million of them involved an active purchase decision on that same day and 2 million of them involved active sell decisions, or about 2.6 percent and 2.2 percent of the time, respectively.

We compute several measures at the position level. First, we construct several different measures of the holding length associated with a given position. Specifically, we consider the length of time (in calendar days) elapsed since the position was first added to the portfolio. In many cases, this measure will be censored because a stock may have been in the portfolio since it was first added to our sample. The average holding length is 485 calendar days (or about 15 months), though this measure is downward-biased. As such, we also examine holding length measures which consider the time elapsed since a stock was most recently bought (or traded). The average position was last purchased about 112 calendar days (a bit less than four months) ago and was last traded about 10 weeks ago. In much of the analysis that follows, we will exclude stocks which were very recently bought to avoid having our results being driven by predictable buying (and lack of selling) behavior as managers split trades over several days while building up positions over time. Second, we compute the portfolio weight as a fraction of market value associated with each position on each date. The average stock has a weight of about 1.2 percent with a standard deviation of 1.6 percent. Inalytics also

Table A.1. Summary statistics

This table reports summary statistics of the analysis dataset for 783 portfolios at various levels of aggregation. The position level summary statistics include various holding lengths, portfolio weights, future return measures and the number of trades (indicator for buy and sell trades). Future returns are reported in percentage points over specified horizons. The fund-level and position-level summary statistics are reported at monthly and daily frequencies, respectively. See Table A.2 and text for additional details on variable construction.

Variable	Count	Mean	Std	25th	50th	75th
Panel A: Fund level Summary						
Heuristics Intensity	0.5M	0.41	0.23	0.29	0.40	0.53
Weekly Fund Flows (%)	1.1M	0.35	5.2	-1.04	0.21	1.79
Concentration (%)	774	2.60	2.59	1.3	2.04	3.02
Large Block Holdings (%)	774	1.09	4.04	0.00	0.00	0.20
Average Size Quintile	783	4.42	0.63	4.23	4.63	4.89
Average Momentum Quintile	783	3.19	0.32	2.97	3.19	3.39
Average Value Quintile	783	2.72	0.50	2.42	2.77	3.02
Panel B : Position Level Summary						
Buying indicator	89.8M	0.03	0.16	0	0	0
Selling indicator	89.8M	0.02	0.15	0	0	0
Holding length since position open (days)	89.8M	484.4	512.9	119	314	679
Holding length since last trade (days)	89.8M	73.36	113.5	10	32	88
Holding length since last buy (days)	89.8M	112.3	152.4	18	57	144
Portfolio weight(%)	89.7M	1.20	1.61	0.24	0.79	1.65
1-day return (%)	82.1M	0.05	4.15	-1.11	0.01	1.17
Future 7-day return (%)	82.9M	0.21	5.83	-2.45	0.18	2.83
Future 28-day return (%)	82.8M	0.78	11.04	-4.63	0.81	6.18
Future 90-day return (%)	82.6M	2.56	20.16	-7.71	2.31	12.30
Future 180-day return (%)	81.5M	5.32	30.51	-10.46	4.16	18.88
Future 270-day return (%)	80.3M	7.87	38.54	-13.10	5.56	24.47
Future 365-day return (%)	78.9M	10.37	44.84	-15.08	7.24	29.73
Earnings announcement day indicator	49.3M	0.01	0.08	0	0	0
Active share	89.8M	0.86	1.27	0.11	0.55	1.28

provides us with a measure of active share.

We also compute a number of backward or forward-looking return measures at the position level over various horizons, both overall and relative to the benchmark return. With the exception of 1-day measures (which refer to the prior trading day), we calculate horizons in calendar days.⁵⁸ For brevity, we only report summary statistics

⁵⁸This choice is, in part, motivated by the fact that trading calendars differ slightly across exchanges. We take a number of precautions to reduce the potential influence of measurement errors in prices,

Table A.2. Summary of characteristics

This table describes how we construct several characteristics for use in our analysis.

Characteristics	Sorting	Construction
Cumulative Returns capped at K-days	Within Fund-date across stocks	$r_{s,f,t}^{cum} = \prod_{i=t-\min\{K,d\}}^{i=t} (1 + r_{s,f,t}) - 1$, where d is the time since a position opens.
Position past k day returns	Within Fund-date across stocks	$r_{s,f,t}^{past\ k} = \prod_{i=t-k}^{i=t-1} (1 + r_{s,f,t}) - 1$.
Fund past k day returns	Across funds on daily basis	$r_{f,t}^k = \prod_{i=t-k-1}^{i=t-1} (1 + r_{f,t}) - 1$.
Heuristics Intensity	Across/Within funds on weekly/monthly basis	$\frac{\text{Total \# of Positions sold in Bin 1 or Bin 6 of past returns}}{\text{Total \# of Positions Sold}}$.
Position Size	Within Fund-date across stocks	$PositionSize_{s,f,t} = \frac{Quantity_{s,f,t}^{beginning\ t} \times P_{s,f,t}}{Fund\ AUM_{s,f,t}}$.
Active share	Within Fund-date across stocks	Position size - weight in client-designated benchmark.
Net Buy	Within funds on weekly basis	# of stocks bought - # of stocks sold.
Monthly Turnover	Across funds on monthly basis	$turnover_{f,m} = \frac{\min\{total\ MarketValue_{f,m}^{buy}, total\ MarketValue_{f,m}^{sell}\}}{MarketValue_{f,m}}$.
Holding length last buy	Within Fund-date across stocks	# of trading days from last day on which a position was bought
Fund Flows	Within Fund across days	$flow_{f,t,k} = \frac{AUM_{f,t} - AUM_{f,t-k} r_{f,t}^k}{AUM_{f,t-k}}$
Position Size	Within Fund within date	$\frac{Shares_{f,i,t} * Price_{i,t}}{AUM_{f,t}}$
Trade Size	Within Fund within date	$\frac{ Shares_{f,i,t} * Price_{i,t} - Shares_{f,i,t-1} * Price_{i,t} }{AUM_{f,t}}$
Value Quintile	By security within year	Quintile of Book to Market ratio relative to other securities in Worldscope
Size Quintile	By security within year	Quintile market capitalization relative to other securities in Worldscope
Momentum Quintile	By security within year	Quintile lagged annual return relative to other securities in Worldscope
Fund Characteristic Quintile	By Fund within date	Weighted average quintile of their holdings for the characteristic of interest (Value, Size, Momentum)
% Large Block Holdings	Calculated by fund/date averaged by fund	$\frac{\sum w_{i,f,t} I(w_{i,f,t} > 0.05)}{\sum w_{i,f,t}}$
% Long Term Holdings	Calculated by fund/date averaged by fund	% of fund AUM held at least 2 years
Concentration	Calculated by fund/date averaged by fund	Average weight of a security in the portfolio
Amihud Liquidity Measure	Calculated by security for each date	$\frac{DollarVolume}{ Past30DayReturn }$

including winsorizing 0.1 percent of returns in either tail by date. These steps are discussed at greater length in the Appendix A.1.

for forward-looking returns that are not adjusted for the benchmark. Volatilities of individual stocks are quite large, with a standard deviation of 45 percent at a 1 year horizon. As we discuss further below, we also consider several measures of prior position performance that are computed using periods of time which depend on holding period length.

There are several characteristics we calculate at the position level in order to explore differences between the investment approaches of different managers. We calculate the percentage of holdings which are long-term (held more than 2 years), and which are held in large blocks ($\geq 5\%$ of the portfolio AUM), and portfolio concentration (average weight of a security in a portfolio). Following the approach of [Daniel et al. \(1997\)](#), we also compute the value, size, and momentum quintiles for securities based relative to the global sample by looking each year where the security’s characteristics fall in rank order relative to the rest of the Worldscope database (results are insensitive to computing these breakpoints at the nation, region, or aggregate level). For the purposes of making DGTW adjustments and/or forming counterfactual portfolios matched on characteristics, these security quintiles are computed annually using Worldscope characteristics (see Appendix [A.3](#)). For purposes of comparing across managers in the heterogeneity analysis in Figure [2](#), we simply average these assigned characteristics at the portfolio level across all positions held.

A.3 Adjusting performance measures for risk and other systemic factors

In this section, we provide additional details about how we construct counterfactuals to form “factor-neutral” portfolios. Specifically, we estimate stock-level exposures to the Fama-French/Carhart 4 factors using data from prior to the trade, then use these estimates to adjust our long short portfolios for ex-ante differences in these exposures.⁵⁹ For each stock-date, we subtract off the inner product of factor loadings and factor realizations, so

$$R_{i,t}^{FN} \equiv R_{i,t} - \Lambda'_{i,q(t)-1} F_t,$$

where $R_{i,t}$ is stock i ’s excess return on date t and F_t is a (4×1) vector of factor realizations.

⁵⁹The four factors are the market excess return, the Fama-French ([2012](#)) international size and value factors, as well as the Carhart momentum factor. As above, we compute loadings using data for the global factors from Ken French’s website.

$A_{i,q(t)-1}$ is a (4×1) vector of factor loadings which are estimated 1 year of daily data using data up to the end of the previous calendar quarter with [Dimson \(1979\)](#) adjustments for asynchronous trading. This adjustment includes one lead and lag for each factor, summing over the contemporaneous, lead, and lag betas. $R_{i,t}^{FN}$ captures return of a self-financing portfolio which, if factor loadings are estimated correctly and are stable, has zero exposure to the priced risk factors on each date. Thus, if the asset pricing model holds, all $R_{i,t}$ should earn zero excess return in expectation, and, accordingly, randomly sold portfolios should have the same factor-neutral returns period-by-period as actual stocks sold. Next, we compute value-added as before, by compounding factor neutral returns then compare cumulative factor-neutral returns of stocks traded with the average of cumulative factor-neutral returns of stocks held.

In order to test whether exposures to known characteristics are driving our results, rather than only correcting for factor exposures, we also use the Worldscope data to construct characteristic-matched portfolios. To do this, we take the annual panel of characteristics for the securities, and construct 5 equal buckets along Book-to-Market (value), Market Capitalization (size), and the 1 year return for the previous year (momentum). This follows the methodology of ([Daniel et al. 1997](#)). We calculate portfolio returns for each portfolio in the 5x5x5 sort, and merge this data in to our panel of security returns (matching on the same quintile of characteristics). This allows us to calculate the excess returns of a security over the matched characteristic portfolio on a daily basis (referred to in the text as DGTW-adjusted returns or characteristic selectivity). The DGTW adjustment calculates the excess period return of the security over the matched portfolio; in our case we compute the excess return on a daily basis. We then take the excess returns for each security and aggregate with the same methodology as our other counterfactuals, grouping into equal-weight portfolios based on whether security was bought, sold, or neither on a given day.

A.4 Standard error computation for counterfactual return measures

In sections [3.2](#), [4](#), and [5](#), we estimate a number of average performance measures which are computed as a weighted average of cumulative long-short portfolio returns ($R_{buy} - R_{hold}$ and $R_{hold} - R_{sell}$). In this section, we describe our methodology for estimating double-clustered heteroskedasticity and autocorrelation robust standard errors associated with these mean calculations. We use a clustered standard error estimator which allows for autocorrelation and heteroskedasticity at the fund level following

a similar approach to autocorrelation as [Hansen and Hodrick \(1980\)](#), who propose a standard error estimator for a general moving average process with a finite number of lags.

For each fund, let $\tau \in \{1, \dots, T_i\}$ index dates for which we have return measures for fund i , and $w_{i,\tau}$ be the weight associated with the τ^{th} observation for fund i and $t_i(\tau)$ be the mapping from the index τ to calendar time t for fund i . Our weighted mean calculation takes the following form for various choices of horizon H :

$$\bar{R} = \sum_{i=1}^N \sum_{\tau=1}^{T_i} w_{i,\tau} R_{i,t_i(\tau):t_i(\tau)+H}, \quad (1)$$

where $\sum_{i=1}^N \sum_{\tau=1}^{T_i} w_{i,\tau} = 1$ and $R_{i,t_i(\tau):t_i(\tau)+H}$ is the cumulative return measure for the associated portfolio for fund i from $t_i(\tau)$ to $t_i(\tau) + H$. Note that (1) also corresponds to the OLS formula for the following regression:

$$\sqrt{w_{i,\tau}} R_{i,t_i(\tau):t_i(\tau)+H} = \sqrt{w_{i,\tau}} \left(\bar{R} + \varepsilon_{i,t_i(\tau):t_i(\tau)+H} \right). \quad (2)$$

To implement a clustered estimator which accounts for both autocorrelation and cross-fund correlation, we also need to estimate

$$Cov[\varepsilon_{i,t(\tau_1):t(\tau_1)+H} \cdot \varepsilon_{i,t(\tau_2):t(\tau_2)+H}] = Corr[\varepsilon_{i,t(\tau_1):t(\tau_1)+H} \cdot \varepsilon_{i,t(\tau_2):t(\tau_2)+H}] \cdot Var[\varepsilon_{i,t:t+H}] \quad (3)$$

and

$$Cov[\varepsilon_{i,t(\tau_1):t(\tau_1)+H} \cdot \varepsilon_{j,t(\tau_2):t(\tau_2)+H}] = Corr[\varepsilon_{i,t(\tau_1):t(\tau_1)+H} \cdot \varepsilon_{j,t(\tau_2):t(\tau_2)+H}] \cdot \sqrt{Var[\varepsilon_{i,t:t+H}] Var[\varepsilon_{j,t:t+H}]}, \quad (4)$$

for cases when $i \neq j$. We assume that $\varepsilon_{i,t_i(\tau):t_i(\tau)+H}$ has mean zero in the cross-section and in the time series, but $Cov[\varepsilon_{i,t_i(\tau_1):t_i(\tau_1)+H}, \varepsilon_{i,t_i(\tau_2):t_i(\tau_2)+H}]$ may be larger than zero for $|\tau_1 - \tau_2| < H$ due to the overlapping structure in calendar time, and cross-portfolio correlations may be non-zero at each point in the time series for lags $< H$. Then, a general formula for the clustered standard error for equation (2) in this case (see, e.g.,

(Cameron and Miller 2015) is

$$V_{clu}[\bar{r}] = \frac{\sum_{i=1}^N \sum_{j=1}^N \sum_{\tau_1=1}^{T_i} \sum_{\tau_2=1}^{T_j} w_{i,\tau_1} w_{j,\tau_2} E[\varepsilon_{i,t(\tau_1):t(\tau_1)+H} \cdot \varepsilon_{j,t(\tau_2):t(\tau_2)+H}] \mathbf{1}[t(\tau_1), t(\tau_2) \text{ in same cluster}]}{\left[\sum_{i=1}^N \sum_{\tau_1=1}^{T_i} w_{i,\tau_1} \right]^2}. \quad (5)$$

Because of the highly unbalanced panel structure of the daily counterfactual returns data, many fund-specific autocorrelation estimates are informed by a very small number of observations. To improve the precision of our estimates, we pool information across funds to obtain more accurate estimates of autocorrelations while still allowing for heteroskedasticity. Exploiting the identity

$$Cov[\varepsilon_{i,t(\tau_1):t(\tau_1)+H} \cdot \varepsilon_{i,t(\tau_2):t(\tau_2)+H}] = Corr[\varepsilon_{i,t(\tau_1):t(\tau_1)+H} \cdot \varepsilon_{i,t(\tau_2):t(\tau_2)+H}] \cdot Var[\varepsilon_{i,t:t+H}], \quad (6)$$

we apply a common correlation term across funds but allow variances to vary at the fund level. We estimate a different autocorrelation for each lag length, which is measured as the number of weekdays between any two calendar time dates. We do this in three stages. First, at the fund level we estimate $Var[\varepsilon_{i,t(\tau_1):t(\tau_1)+H}]$. Next, on a fund-by-fund basis we estimate $Corr[\varepsilon_{i,t(\tau_1):t(\tau_1)+H} \cdot \varepsilon_{i,t(\tau_2):t(\tau_2)+H}]$ by taking the ratio of the sample mean of the cross terms at a given lag length to our fund-specific variance estimate.

We also need to account for cross-fund correlations, but because calculating it directly requires computationally infeasible combinatorics, we use a leave-one-out calculation, estimating the cross-correlation between a fund's period return and the period/lag returns of the weighted average of the other fund's portfolios. In other words, fixing the lag, we estimate

$$Corr \left[\varepsilon_{i,t(\tau_1):t(\tau_1)+H}, \left(\sum_{j \neq i} \frac{w_{j,t(\tau_2)}}{\sum_{k \neq i} w_{k,t(\tau_2)}} \varepsilon_{j,t(\tau_2):t(\tau_2)+H} \right) \right] \equiv Corr [\varepsilon_{i,t(\tau_1):t(\tau_1)+H}, \varepsilon_{\setminus i,t(\tau_2):t(\tau_2)+H}] \quad (7)$$

for each portfolio i . Then, finally, we aggregate to a single correlation number by taking a weighted average of the individual fund-level autocorrelation and cross correlation estimates and compute the standard error using equations (5-6) with the common correlation and fund-specific variance estimates, respectively.

Inspecting our autocorrelation estimates, it appears that the primary source of au-

tocorrelation in our counterfactual performance measures comes from repeated trading of the same asset. For instance, if a PM sells a stock today, she is also somewhat more likely to sell it tomorrow and in the near future. Since we use cumulative return measures, consecutive trades of the same asset will mechanically be autocorrelated due to a moving average structure in the error term which comes from overlapping calendar time periods for cumulative returns.

Autocorrelation in trading appears to a very high frequency phenomenon. As a result, our estimated autocorrelation for our 1 year counterfactual sell measures is about 50% at a 1 trading day frequency and falls to 23% at a 5 trading day horizon. Within 1 and 3 months, these autocorrelations fall to about 7% and 3%, respectively, and decay rapidly towards zero thereafter. At long horizons, we find little to no evidence of autocorrelation even for periods which overlap in calendar time, as might be expected if stocks bought/sold tend to have different exposures to systematic risk relative to stocks not sold. In light of these estimates, since the computational cost of computing a full set of daily autocorrelations is quite high, we impose an estimate of zero after 100 business days, though overall estimates were quite similar when we allowed all autocorrelations for overlapping calendar time period to be nonzero.

Whereas these autocorrelation adjustments result in a nontrivial increase in estimated standard errors, the adjustment for cross-fund correlations turns out to be quite small. Since our PMs tend to hold concentrated portfolios that depart quite a bit from the benchmark, there is a fairly low (though not zero) overlap between holdings of different portfolios in our sample, and we do not find highly correlated trades in the cross-section. For factor-neutral portfolios, the average cross-fund correlation for the Sell - Hold portfolio when only looking at the same day of trading is 0.01. Over longer horizons, this estimate approaches closer to zero and even flips signs.

The small cross-fund correlations make sense considering the context of the sample—a set of concentrated managers with distinct mandates that are hired to earn excess returns by departing from the benchmark. For example, we observe roughly 18,000 different SEDOLs and 14,000 different ISINs held by the managers in our sample, though each only holds roughly 60 securities at a time. Cross covariances are essentially zero in magnitude, and, if anything, tend to be slightly negative after accounting for the correlation with lagged trades, with an average between-fund correlation of -0.005. In turn, the double clustered standard errors are quite similar and sometimes a bit smaller relative to the single clustered, HAC standard errors, as shown in Table A.3. Note that

Table A.3. Standard Errors single-clustered HAC adjusted vs. double-clustered

Comparison of single-clustered HAC standard errors and double-clustered standard errors for the baseline results presented in Table 3. Double-clustered standard errors are extremely similar to single clustered, and sometimes are smaller due to negative between-fund correlations, as discussed in Appendix A.4.

<i>Factor-Neutral</i>	Panel A: Buy Standard Errors		Panel B: Sell Standard Errors	
	Single-Clustered	Double-Clustered	Single-Clustered	Double-Clustered
Window				
28	0.04	0.04	0.04	0.04
90	0.08	0.09	0.08	0.09
180	0.13	0.15	0.14	0.15
270	0.17	0.20	0.18	0.18
365	0.22	0.24	0.22	0.20
<i>Unadjusted</i>				
Window	Single-Clustered	Double-Clustered	Single-Clustered	Double-Clustered
28	0.04	0.05	0.04	0.04
90	0.08	0.10	0.08	0.10
180	0.13	0.18	0.14	0.17
270	0.17	0.22	0.18	0.20
365	0.22	0.27	0.22	0.23

these results also speak against an explanation based on price impact from coordinated sales across funds. Given the small cross-fund correlations, PMs tend to not sell the same assets at the same time, implying that the price impact channel is unlikely to rationalize our performance results.

As an additional piece of evidence consistent with this argument, Appendix A.10 considers the returns from a set of strategies in which we form calendar time portfolios based on monthly signals which are constructed using our trading data. These portfolios earn quite high Sharpe ratios (though these calculations do not take into account transaction costs associated with implementing such a strategy), in large part because there are substantial benefits from diversifying across different stocks which are sold at given points in time. In addition, these portfolios have very low exposures to systematic factors, consistent with the arguments made throughout the paper that our sample of PMs mostly seek to generate superior performance via selecting individual stocks rather than by betting on systematic risk factors.

A.5 Additional details and robustness checks for performance results

Due to concerns that trading frictions may prevent managers from exiting or entering positions as they see fit (and thus a random security may not be a good counterfactual), we merge in both Worldscope and Datastream to the sample to drop securities which do not meet a minimum threshold for either market capitalization or liquidity. In both cases, we take a measure (either market capitalization or the Amihud liquidity measure), calculate a 5th percentile breakpoint for the universe of securities in either Worldscope (for market cap) or Datastream (for Amihud liquidity) for each year, and merge the relevant sample into our portfolio holdings. We perform tests where we exclude all securities below a certain threshold from all relevant portfolios (buy, sell, and hold) and where we exclude securities only from the hold portfolio (which would be the relevant counterfactual). For one result in the appendix, we also merge NYSE breakpoints taken from Ken French's data library, using the 5th percentile breakpoint from those data.

Because a random counterfactual may not be appropriate (for example, if a portfolio was only comparing securities with similar market capitalizations) we perform a series of matching exercises, where we only compare securities to similar securities matched on characteristics. In order to construct those counterfactuals, we construct quintiles for each portfolio-date along some characteristic of interest, and use the portion of the portfolio which is held (and not traded) within that same quintile as a counterfactual. When aggregating across these quintiles, we weight the average by the number of buy or sell trading days within a specific quintile. Quintiles are calculated based on the source data, grouping annually. For example, Worldscope provides an annual panel of characteristics, and we compute quintiles by year before merging with our sample.

Despite the relatively small AUMs (compared to large mutual funds), one may be concerned that funds may have the potential for price impact when buying or selling. To address this concern, we have constructed a number of counterfactuals that start a certain number of days after the trade. In the paper we report several of these specifications, which are all constructed in the same way other than the number of days omitted. While we only report results for one week for brevity, omitting the first few days post-trade has the tendency to make selling performance look worse, not better, which is the opposite of what we would predict from a price impact mechanism.

This appendix also reports several additional robustness exercises that address price

Table A.4. Post-trade unadjusted returns relative to counterfactual, additional robustness checks

This table presents the average value added measures (post-trade unadjusted returns relative to a random sell counterfactual) for buy and sell trades. We show all unadjusted versions of the specifications in Table 3 which were only presented in factor neutral returns.

Performance Measure	Panel A: Buy						Panel B: Sell					
	28	90	180	270	365		28	90	180	270	365	
Drop 5% smallest securities	0.19 (0.05)	0.41 (0.10)	0.46 (0.17)	0.56 (0.21)	0.60 (0.25)		-0.05 (0.04)	-0.23 (0.10)	-0.76 (0.15)	-0.75 (0.19)	-0.85 (0.24)	
Drop 5% most illiquid securities	0.41 (0.05)	0.70 (0.11)	0.92 (0.19)	1.08 (0.24)	0.26 (0.31)		0.03 (0.04)	-0.15 (0.09)	-0.37 (0.16)	-0.42 (0.20)	-0.60 (0.24)	
Value-weighted	0.36 (0.05)	0.75 (0.15)	1.06 (0.30)	1.32 (0.39)	1.56 (0.47)		-0.01 (0.05)	-0.27 (0.13)	-0.62 (0.26)	-0.85 (0.34)	-1.00 (0.39)	
Drop most illiquid quintile by portfolio	0.35 (0.05)	0.61 (0.10)	0.93 (0.17)	1.14 (0.23)	1.33 (0.30)		0.05 (0.04)	-0.17 (0.09)	-0.35 (0.14)	-0.36 (0.18)	-0.50 (0.22)	
Drop shortest quintile of holding length	0.41 (0.04)	0.77 (0.10)	1.07 (0.18)	1.27 (0.24)	1.49 (0.28)		0.03 (0.04)	-0.18 (0.10)	-0.55 (0.17)	-0.70 (0.20)	-0.81 (0.23)	
Drop week post trade, Unadjusted Returns	0.21 (0.04)	0.52 (0.09)	0.74 (0.16)	0.88 (0.21)	1.05 (0.25)		-0.03 (0.04)	-0.21 (0.09)	-0.53 (0.17)	-0.71 (0.21)	-0.82 (0.24)	
<i>Match counterfactual on</i>												
Market capitalization quintile	0.33 (0.05)	0.50 (0.10)	0.54 (0.18)	0.61 (0.23)	0.67 (0.28)		-0.16 (0.04)	-0.41 (0.09)	-0.44 (0.14)	-0.51 (0.18)	-0.62 (0.22)	
Liquidity quintile	0.38 (0.05)	0.65 (0.11)	0.86 (0.19)	1.00 (0.25)	1.19 (0.31)		0.05 (0.05)	-0.18 (0.10)	-0.46 (0.18)	-0.49 (0.22)	-0.63 (0.25)	
Idiosyncratic volatility quintile	0.39 (0.04)	0.68 (0.09)	0.90 (0.14)	1.01 (0.19)	1.02 (0.22)		0.00 (0.04)	-0.23 (0.09)	-0.56 (0.15)	-0.76 (0.18)	-0.92 (0.21)	
Position size quintile	0.32 (0.05)	0.51 (0.09)	0.57 (0.13)	0.71 (0.18)	0.94 (0.24)		0.03 (0.04)	-0.17 (0.09)	-0.56 (0.15)	-0.79 (0.17)	-0.92 (0.20)	
Previous quarter's benchmark adjusted return	0.38 (0.04)	0.73 (0.09)	0.99 (0.13)	1.23 (0.17)	1.45 (0.23)		0.07 (0.04)	-0.08 (0.08)	-0.33 (0.13)	-0.49 (0.17)	-0.66 (0.21)	

Table A.5. Post-trade factor-neutral returns relative to counterfactual, additional robustness checks

This table presents the average value added measures (post-trade returns relative to a random sell counterfactual) for buy and sell trades. We show several additional robustness checks on a factor-neutral basis.

Performance Measure	Panel A: Buy					Panel B: Sell				
	28	90	180	270	365	28	90	180	270	365
Drop 5% smallest, NYSE breakpoints	0.16 (0.05)	0.33 (0.10)	0.25 (0.17)	0.44 (0.21)	0.48 (0.26)	0.01 (0.04)	-0.12 (0.10)	-0.55 (0.16)	-0.49 (0.20)	-0.69 (0.23)
Drop 20% smallest, NYSE breakpoints	0.28 (0.04)	0.49 (0.09)	0.59 (0.15)	0.61 (0.19)	0.58 (0.25)	0.00 (0.02)	-0.24 (0.09)	-0.55 (0.12)	-0.62 (0.17)	-0.76 (0.20)
Drop most illiquid quintile by portfolio	0.41 (0.05)	0.70 (0.11)	0.92 (0.19)	1.08 (0.24)	1.26 (0.31)	0.03 (0.04)	-0.15 (0.09)	-0.38 (0.16)	-0.42 (0.20)	-0.60 (0.24)
Drop 5% most illiquid from counterfactual	0.35 (0.05)	0.61 (0.09)	0.89 (0.15)	1.05 (0.20)	1.29 (0.26)	0.02 (0.05)	-0.24 (0.10)	-0.49 (0.16)	-0.55 (0.20)	-0.71 (0.22)
Drop 5% smallest from counterfactual	0.34 (0.04)	0.57 (0.09)	0.73 (0.16)	0.87 (0.21)	0.91 (0.26)	0.00 (0.04)	-0.30 (0.11)	-0.69 (0.16)	-0.88 (0.20)	-1.03 (0.23)
Omit largest holdings quintile	0.31 (0.06)	0.58 (0.13)	0.87 (0.27)	1.11 (0.38)	1.37 (0.46)	-0.05 (0.06)	-0.42 (0.18)	-0.75 (0.37)	-0.98 (0.44)	-1.17 (0.45)
<i>Match counterfactual on</i>										
Market capitalization quintile	0.28 (0.04)	0.37 (0.09)	0.44 (0.13)	0.50 (0.16)	0.51 (0.19)	0.00 (0.05)	-0.22 (0.09)	-0.47 (0.16)	-0.48 (0.20)	-0.58 (0.26)
Book to market quintile	0.35 (0.05)	0.56 (0.09)	0.66 (0.14)	0.78 (0.19)	0.92 (0.23)	0.03 (0.04)	-0.13 (0.10)	-0.49 (0.15)	-0.45 (0.18)	-0.59 (0.22)
Momentum quintile	0.32 (0.05)	0.49 (0.09)	0.56 (0.15)	0.62 (0.19)	0.73 (0.24)	0.01 (0.04)	-0.20 (0.09)	-0.54 (0.15)	-0.66 (0.18)	-0.68 (0.20)
Liquidity quintile	0.32 (0.05)	0.53 (0.10)	0.82 (0.17)	0.98 (0.22)	1.19 (0.28)	0.04 (0.05)	-0.22 (0.10)	-0.43 (0.16)	-0.48 (0.20)	-0.56 (0.23)

impacts and illiquidity. First, for each of the specifications in Table 3 that only reported results with factor neutral returns, A.4 reports the same specifications with unadjusted returns. In A.4 and A.5 we also perform a wide variety of additional tests. The first batch are focused on issues of liquidity, e.g. that managers are unable to sell securities from the counterfactual due to liquidity concerns, or that there may be large price impacts in illiquid markets. In addition to the unadjusted version of the tests reported in Table 3, we conduct several additional tests. First, we use NYSE 5% breakpoints (rather than the Worldscope 5% breakpoints), which have a higher marketcap threshold due to the fact that NYSE stocks are larger than most of the emerging markets equities in our sample. We also include more aggressive versions of the analysis which drop the 20% least liquid portion of a given portfolio's holdings on a given day, finding similar results as our main specification.

In addition, because the portfolio managers *do* choose to trade the securities in the buy and sell portfolios, the liquidity concerns for the unobserved trades must be in the set of holdings that were not traded on a given day. It is plausible that managers have skill when choosing to trade illiquid securities, or are intentionally providing liquidity to the market at a specific time to generate excess returns. Thus, we also perform tests where we only exclude small and illiquid securities from the counterfactual portfolio, finding that the results are similar to the original specification, often with larger magnitudes than the baseline result.

It may also be the case that portfolio managers have specific style tilts, tight tracking error budgets, or position size limits. We perform additional tests to make sure that those potential issues are not driving our result. Rather than assigning a random counterfactual, we condition on the counterfactual matching the quintile of the sold security along the respective characteristics, including position size, market capitalization, book to market ratio, momentum, and liquidity. We cannot jointly test these because the dimensionality of a tri-variate sort quickly exceeds the average number of securities held by a given fund, but each univariate matched counterfactual has similar results to the baseline. The fact that, after accounting for factors at the security level *and* matching the counterfactual on traditional style characteristics, we still observe the same performance results for both buying and selling should also assuage concerns that any systematic return premium is driving our findings.

Finally, we directly address potential position size limits by excluding the largest quintile of holdings in each portfolio from the buy and sell portfolios. If sells that lacked

proper attention were driven by automatic rebalancing based on upper limits for a given position, we would expect that underperformance to be driven by “forced trades” of a PM’s largest positions, measured as a proportion of their AUM on the day prior to the trade. We find that, if anything, selling underperformance becomes more pronounced when we exclude the top 20% of a portfolio’s securities. This implies that, consistent with our attention mechanism, sales of a PM’s largest positions (which are likely to be attended to) tend to look better sells relative to the average sell. Accordingly, when we drop the largest quintile of position size from the analysis (trade and counterfactual portfolios), we see that average selling performance measures deteriorate even further.

We can also conduct a simple model-free diagnostic which simultaneously addresses two potential alternative explanations for our results mentioned in the main text. First, results could be driven by common exposures of stocks traded to common systematic, priced factors which were omitted from our explicit risk-adjustment procedures. Second, tracking error constraints might imply that a stock sold might need to share common systematic risk exposures with a stock purchased (e.g., to hold constant exposure to a given industry or region within the portfolio), leading the stocks sold to potentially share exposures to these factors (which subsequently outperform) as stocks purchased. We can address both of these potential arguments to some extent without needing to take a stand on what these specific systematic factors might be. The basic logic is as follows: if stocks purchased and stocks sold shared a common exposure to an unobserved risk factor which was not also shared by stocks held, we would expect to see that stocks bought and sold contemporaneously in time to have returns that are fairly highly correlated with one another, even after adjusting for exposures to common risk factors that appear in our factor-neutral portfolios.

To get at this question, we compute the correlation between R_{buy} and $R_{sell} - R_{hold}$ for stocks that occur on the same day or within a few trading days. Intuitively, if the stocks sold tend to share common systematic risks with stocks bought, we would expect to see a large positive correlation between R_{buy} and R_{sell} —a much larger one than between R_{hold} and R_{sell} . To get an idea of some magnitudes that we might expect to see, (Campbell, Lettau, Malkiel, and Xu 2001) decompose stock returns into market, industry-specific, and firm-specific components, and find that the variance of the industry-specific component (after residualizing with respect to the market and using the fairly coarse Fama-French 49 industry classification) is around 0.001 at a monthly frequency. Likewise, the firm-specific component has a variance of around

Table A.6. Post-trade returns relative to counterfactual for trades with $\geq 50\%$ change in weights

This table presents the average value added measures for large trades ($\geq 50\%$ change in the weight of a position) under two measures of returns 1) raw returns and 2) factor-neutral returns. Double-clustered standard errors, computed using the method described in the section A.4, are reported in parentheses.

	Panel A: Large Buys					Panel B: Large Sells				
	28	90	180	270	365	28	90	180	270	365
Unadjusted	0.60 (0.07)	1.04 (0.17)	1.28 (0.29)	1.45 (0.35)	1.72 (0.42)	-0.43 (0.09)	-0.67 (0.27)	-1.05 (0.50)	-1.25 (0.75)	-1.27 (0.86)
Factor-Neutral	0.54 (0.05)	0.81 (0.09)	1.04 (0.19)	1.18 (0.25)	1.44 (0.31)	-0.45 (0.08)	-0.70 (0.22)	-1.09 (0.48)	-1.19 (0.65)	-1.11 (0.70)

0.005. This would imply that the R^2 of a regression of a firm stock return (after absorbing the market) on an industry-specific portfolio would be in the neighborhood of 0.17, or, in other words, the correlation between the firm-specific component and the industry-specific component is around 40%.

In our data, the contemporaneous correlation of $R_{buy} - R_{hold}$ and $R_{sell} - R_{hold}$ when buys and sells happen on the same day is between 1.5% and 2.5%, depending on the future return horizon we look at, implying an R^2 of at most about 0.0006. This suggests that the stocks bought have essentially the same covariance with stocks sold as stocks held (which could have been sold instead). These correlations are essentially zero for trades occurring one or more trading days apart from one another. Such magnitudes are far too small to support an explanation based on exposures to unobserved systematic factors.

The results thus far have examined performance of all buying and selling decisions together. However, both buys and sells differ in the extent to which they add or subtract from the portfolio. Some buys add a little bit to an existing position while others introduce a substantial amount of shares or start a whole new position in the portfolio; similarly, some sells cut a bit from existing positions while others unload substantial shares or cut the asset altogether. We look separately at decisions which change the weight of a position by 50 percent or more, relative to the current weight of a position in the portfolio. For example, a sell which changes the relative weight of a position from 10% to 5% or a buy which changes a position's weights from 20% to 40% would fall into this category. We also include new purchases and full liquidations in this

category. Table A.6 above presents these results relative to the same counterfactual used in Table 3. The same pattern of results is observed.⁶⁰

A.6 Additional Institutional Context: Reporting of Buy vs Sell Performance

In this section, we discuss the common approaches to reporting performance and the extent to which they facilitate the identification of value-added from buying and selling decisions. PMs and their clients generally use Brinson-style attribution to report and monitor their portfolio performance. This method looks at sector/country/style characteristics and measures the portfolio's over- or under-weights relative to a benchmark. The Brinson-Fachler model for attribution, which is the industry standard method of performing this sort of analysis, breaks down the excess return of a portfolio over its benchmark into an "allocation" effect, a "selection" effect, and an "interaction" effect.

The allocation effect is constructed by multiplying the extent to which a portfolio overweights a given sector (or other category such as country) by the extent to which the benchmark return for that sector outperformed the overall benchmark return. This is carried out separately for each sector, and may be added for all sectors to give an overall allocation effect. Having a separate allocation and selection effect separates decisions related to which sector the PM chose to invest in from the actual names picked within each sector.

The selection effect assumes the sector weights of the benchmark, but uses the aggregate portfolio returns in each sector in place of the benchmark returns, and then looks at the difference. This component is meant to identify poor security selection. However, because this uses aggregate returns, the only context in which it could identify selection underperformance is if the portfolio, on average, holds worse securities than the benchmark does, within any given sector. In other words, even if recent sell trades hurt performance by prematurely reducing exposure to a security that outperformed, these attribution measures would still give the manager credit for positive security selection because they are based on portfolio weights, not changes in weights.

As a consequence of this reporting standard, even if a portfolio is underperforming its benchmark after adjusting for industry weights, the emphasis of such attribution is

⁶⁰Note that the directionally greater performance of large relative buys compared to other buy trades is consistent with PMs trading on information when opening or increasing a position by a significant amount.

the current holdings. The only way that costs from selling could be easily isolated is if the portfolio previously had a large position in a security that also had a large weight in the benchmark, and also that such a security was one of the largest contributors to the benchmark return. In such a case, we believe it to be plausible that a manager/client would take another look at their selling decision, especially because underweighting a security which has a large position in the benchmark is typically an active choice. However, because of the nature of the management style (concentrated portfolios that depart from benchmark), most of the securities that are sold are not large positions in the benchmark.

Given the emphasis on performance of existing holdings, this reporting standard makes it much more difficult to identify underperformance in sales than in buys, which naturally show up as current holdings. In turn, managers are less likely to notice that they are using costly heuristics when selling, which perpetuates underperformance relative to counterfactual measures. Noticing these costs is especially challenging in light of the horizon over which underperformance takes place. While we find that these potentially inattentive selling decisions have neutral performance over short horizons (e.g., less than 1 month), they substantially underperform over longer ones. This added delay between the decision and the realization of the outcome further adds to the difficulty of learning from prior decisions.

A.7 Additional results and information related to trading probabilities

This section provides additional details about our analysis in section 4.4.1. First, we discuss how the bins are constructed, then report additional results related to the probability of making buy and sell decisions, mainly conditional specifications which separate out large trades, sort on position size and holding length, or condition on a day being a “trading day.”

On each trading date, we sort stocks into N_{bin} bins using these relative rankings. We always choose an even number of bins and always set the breakpoint between bins $N_{bin}/2$ and $N_{bin}/2 + 1$ equal to zero. This ensures that all stocks in bins $N_{bin}/2$ have declined relative to the benchmark. We choose all remaining breakpoints so that (ignoring issues related to discreteness) there are equal numbers of stocks in bins $1, \dots, N_{bin}/2$ and bins $N_{bin}/2 + 1, \dots, N_{bin}$. As a baseline, we consider $N_{bin} = 20$. Some specifications collapse across bins to fit more conveniently in tabular format—

the results are always robust to the number of bins (N_{bin}) considered.

Throughout our analysis of trading probabilities, we make one substantive restriction on the sample of stocks which are under consideration. In predicting the probability that a manager will add to/reduce an existing position, we exclude stocks that were bought in the very recent past. Specifically, we sort positions based on the holding length since the last buy trade and exclude the shortest 20% (shortest time elapsed since last purchase) from our calculations. We do this to avoid a fairly mechanical relationship between our prior return measure, which has a variance that shrinks with the holding period, and the probability of buying/selling that can be generated if managers build up positions by splitting buy trades over short windows of time.⁶¹ Such trades likely originate from a single purchase decision being executed over time, and so we construct our measures to treat them as such. Further, to ensure meaningful distinctions between bins, we exclude fund-dates which include fewer than 40 stocks in the portfolio, though results for predicted selling probabilities do not meaningfully change without such a restriction.

In the main text, we focus on PMs' decisions to purchase/sell existing holdings. Since PMs can add new positions to their portfolios, it is highly likely that they have a wider consideration set than existing holdings which is relevant from their buying decisions. Here, we describe a complementary approach which also allows for new stocks to be added to the portfolio in addition to current holdings. This approach includes all purchase decisions—including the opening of brand new positions—and calculates relative prior returns by broadening the consideration set to assets that are likely being considered for purchase. For this wider consideration set approach, we use the prior 1 quarter benchmark-adjusted return. Specifically, because our dataset contains not only current and past holdings for each PM but future holdings as well, we can include assets that the PM is likely considering by looking at what he ended up buying within 12 months of the current date. We include those assets in the portfolio when computing the prior return bins to examine whether new positions are more or less likely to be bought depending on prior returns relative to the larger consideration set.

In Table A.7, Panel A replicates the buying decisions presented in Figure 3 but

⁶¹This phenomenon mechanically tends to increase the likelihood that positions with non-extreme returns are bought and decrease the likelihood that they are sold, since a manager is unlikely to sell an asset immediately after or while actively building a position in it.

Table A.7. Probability of buying and selling: larger consideration set and large trades

This table presents the probability of buying for security in the investors holdings and consideration sets, as well as the probability of trades with a $\geq 50\%$ change in weights as a function of prior returns. Panel A presents reports differences in probabilities, in percentage points, of buying selling by bins of past benchmark-adjusted returns for stocks in the larger consideration set (current holdings plus stocks added to the portfolio in the next year) by 20 bins of positions past benchmark-adjusted returns. The baseline probability of trading a stock in the omitted category, bin 10, is reported in the rightmost column. The first row applies the same filters as Figure 2, while the second row considers a wider set of stocks following the approach described in the main text. Panel B presents the relative probability of large buys and sells by 20 bins of past positions benchmark-adjusted returns capped at 90 days.

Panel A: Buying probability by bins of prior 1 quarter benchmark-adjusted returns

Buying prob for	Differences relative to bin 10 for prior return bins								Baseline prob (bin 10)
	1	2	3-5	6-9	11-15	16-18	19	20	
Current holdings	0.03%	0.05%	0.03%	0.00%	0.02%	0.00%	-0.02%	-0.06%	1.10%
Consideration set	0.09%	0.07%	0.05%	0.00%	0.00%	0.07%	0.10%	0.06%	1.62%

Panel B: Trading probability by bins of prior benchmark-adjusted returns, capped 1 quarter

Trading Probability	Differences relative to bin 10 for prior return bins								Baseline prob (bin 10)
	1	2	3-5	6-9	11-15	16-18	19	20	
Large Buys	0.02%	0.02%	0.01%	0.02%	0.01%	0.01%	0.01%	0.01%	0.16%
Large Sells	0.38%	0.27%	0.21%	0.07%	0.01%	0.01%	0.03%	0.12%	0.37%

includes new buys using our second approach which expands the consideration set. Specifically, we report differences in probabilities relative to a baseline category (bin 10, stocks which barely underperformed the benchmark) of buying across categories of prior returns. For ease of comparison, the top row reports our estimate from Panel B of Figure 3, which uses the 1 quarter prior benchmark-adjusted return measure as the sorting variable. We average probabilities across several intermediate bins for brevity, and report the baseline probability associated with the omitted category in the final column. Then, the second row uses the same sorting variable but also includes stocks in the broader consideration set (as defined above) and eliminates our restriction which excluded stocks in the bottom bin of holding length since last buy. We see that the probabilities of purchasing an asset remain quite flat with respect to prior returns.

Panel B of Table A.7 depicts the propensity to engage in large transactions—selling more than 50 percent of an asset—as a function of prior returns. We see a similar U-shape emerge as when we consider the all sales together. Again, this pattern is

not matched for the probability of engaging in large purchases, which remain quite flat across bins of prior returns. Together, these results demonstrate that we can predict selling decisions based on observables from the PM's current holdings with some confidence; in contrast, these observables—nor any others that we have considered—do not predict buying decisions.

We also examine whether our observed pattern can be explained by other variables which could be correlated with our prior return measures: holding length and position size. In the former case, as discussed above, positions which have only been held for a short period of time will to have less dispersion in returns and also may be more likely to be bought and less likely to be sold. In the latter, even if initial positions all begin at the same size, extreme returns could lead them away from target weights, and rebalancing motives could drive managers to sell positions with extreme positive returns that have become too large.⁶²

As in Table 3, we assign each stock into one of 20 bins based on prior returns and the other sorting variable, respectively. Since the break points used for the second characteristic are the same regardless of the bin associated with the first characteristic, there will be unequal numbers of observations in each bin. We then report the buying (top panel) or selling (bottom panel) probabilities within each group relative to the middle, least extreme bin (bin 10). As in Figure A.7, we average across several intermediate categories and separately report the probability of trading for the omitted category.

Panel A of Table A.8 double sorts on six bins based on time elapsed since last buy (the variable we filter on) and prior returns. For this analysis only, we do not discard any stocks from the analysis based on the holding period measure. One can observe the mechanical patterns discussed in Section 4.4.1 when looking at the buying probabilities of assets in the bin with the shortest holding length; buying probabilities are quite flat in prior returns for longer holding periods. In contrast, assets in extreme bins are much more likely to be sold across all holding lengths.

As shown in Panel A of Table A.8, we observe that selling probabilities feature a

⁶²Note, however, that similar logic would potentially imply that we would see less selling of positions that have become small due to portfolio drift, which we do not observe. Also, from the univariate evidence above, we do not see large increases in buying for positions that declined in value, as would be predicted by this channel. In regressions below, we will always include linear and quadratic controls for position size and holding length.

Table A.8. Probability of trading by prior returns and position characteristics

This set of tables reports differences in probabilities, in percentage points, of buying/selling by bins of past benchmark-adjusted returns double sorted with bins of position characteristics holding length and position sizes relative to bin 10 of past benchmark-adjusted returns within each category. The top section of each panel reports relative probabilities of buying and the bottom section reports relative probabilities of selling. Baseline probabilities for the omitted category are reported below. Columns represents different holding lengths in Panel A and position sizes in Panel B. Different bins of past position returns are reported in rows, together with the baseline probability of the omitted category.

Panel A: Holding Length

Trade	Past Return\Holding Length	Shortest	Short	Short-Med	Med-Long	Long	Longest
Buy	1	-2.34	-0.20	-0.15	-0.07	-0.03	0.03
	2	-1.99	-0.05	-0.03	0.02	0.05	0.07
	3-5	-2.08	-0.02	-0.01	0.05	0.04	0.05
	6-9	-1.20	0.05	0.05	0.13	0.07	0.04
	11-15	0.28	-0.03	-0.03	0.01	0.04	0.05
	16-18	-1.59	-0.19	-0.12	-0.03	0.02	0.08
	19	-2.27	-0.33	-0.21	-0.06	0.02	0.09
	20	-2.93	-0.48	-0.29	-0.10	-0.02	0.09
	Baseline: 10	8.78	2.03	1.56	1.22	0.86	0.68
Sell	1	0.45	0.78	1.17	1.14	1.52	1.70
	2	0.31	0.40	0.59	0.64	0.93	1.05
	3-5	0.24	0.24	0.35	0.34	0.42	0.58
	6-9	0.16	0.14	0.19	0.18	0.12	0.16
	11-15	0.07	0.08	0.08	0.02	0.05	0.01
	16-18	0.32	0.27	0.32	0.32	0.35	0.34
	19	0.49	0.47	0.61	0.56	0.63	0.58
	20	0.75	0.94	1.12	1.15	1.25	1.21
	Baseline: 10	1.02	1.63	2.07	2.15	2.17	2.33

Panel B: Position Size

Trade	Past Return\Position Size	Smallest	Small	Small-Med	Med-Large	Large	Largest
Buy	1	-0.21	-0.07	0.11	0.19	0.31	0.49
	2	-0.12	-0.01	0.11	0.15	0.25	0.39
	3-5	-0.11	-0.03	0.05	0.12	0.18	0.27
	6-9	0.05	0.02	0.10	0.06	0.07	0.13
	11-15	0.08	0.01	0.05	0.03	-0.02	-0.02
	16-18	0.02	-0.05	-0.01	-0.02	-0.05	-0.08
	19	-0.07	-0.09	-0.04	-0.04	-0.07	-0.14
	20	-0.10	-0.13	-0.09	-0.09	-0.13	-0.20
	Baseline: 10	0.98	1.07	1.03	1.07	1.17	1.28
Sell	1	1.33	0.90	0.93	1.02	1.02	0.90
	2	0.92	0.54	0.55	0.65	0.64	0.53
	3-5	0.55	0.30	0.31	0.35	0.37	0.30
	6-9	0.17	0.13	0.12	0.12	0.12	0.09
	11-15	-0.06	0.04	0.03	0.10	0.13	0.14
	16-18	0.07	0.23	0.30	0.42	0.51	0.59
	19	0.22	0.41	0.51	0.71	0.83	0.96
	20	0.77	0.97	1.13	1.28	1.40	1.66
	Baseline: 10	3.38	1.93	1.81	1.85	1.95	2.18

pronounced U-shape for all position sizes, a pattern that holds robustly within all position size bins. For larger positions, we see some evidence of PMs adding to their biggest positions following losses. However, even within these categories, buying probabilities increase gradually with losses and decrease gradually with gains, whereas corresponding sell probabilities increase much more dramatically for the more extreme return categories. Additionally, the magnitudes for buys are generally much smaller compared to the respective selling probabilities. We discuss position size more in Section 4.1 of the main text.

Table A.9 presents results for the same specification as the linear probability models in Table 5, but conditions on there being at least one trade. In other words, this asks the question, “on days where a PM buys, how do the past extreme returns influence her decision.” For sells, this instead conditions on sells. We find that results are quite similar to those in Table 5, but with larger magnitudes, especially for the selling results.

A.8 Additional results on the attention mechanism

In this section, we present additional evidence that the poor selling skill is driven by a lack of attention to those decisions. Namely we separate out periods where portfolio managers are expected to be stressed (high turnover trades) or which will have less immediate impact on the portfolio weights (small vs. large transactions). We also look at how the heuristic intensity results manifest, presenting evidence that they are less associated with a manager’s average skill, but instead time-variation in their propensity to rely on extreme returns as a heuristic.

Figure A.2 presents results from an alternative approach in which we seek to separate selling decisions which are likely attended to from those that are not. Specifically, we compare the performance of large (top quintile, within portfolio) trades to all other trades. It is likely that portfolio managers think hard and attend to the largest trades, and in turn, we would expect those trades to perform better relative to a counterfactual. By contrast, if we believed the results to primarily be driven by a mechanical price impact, we would expect those same trades to perform the worst. Consistent with the attentional mechanism, all trade size quintiles other than the largest fail to beat the counterfactual. The median-sized trades underperform the counterfactual substantially.

Periods of high turnover are likely to be those during which a portfolio manager’s

Table A.9. Probability of trading based on prior returns, conditional on buy or sell

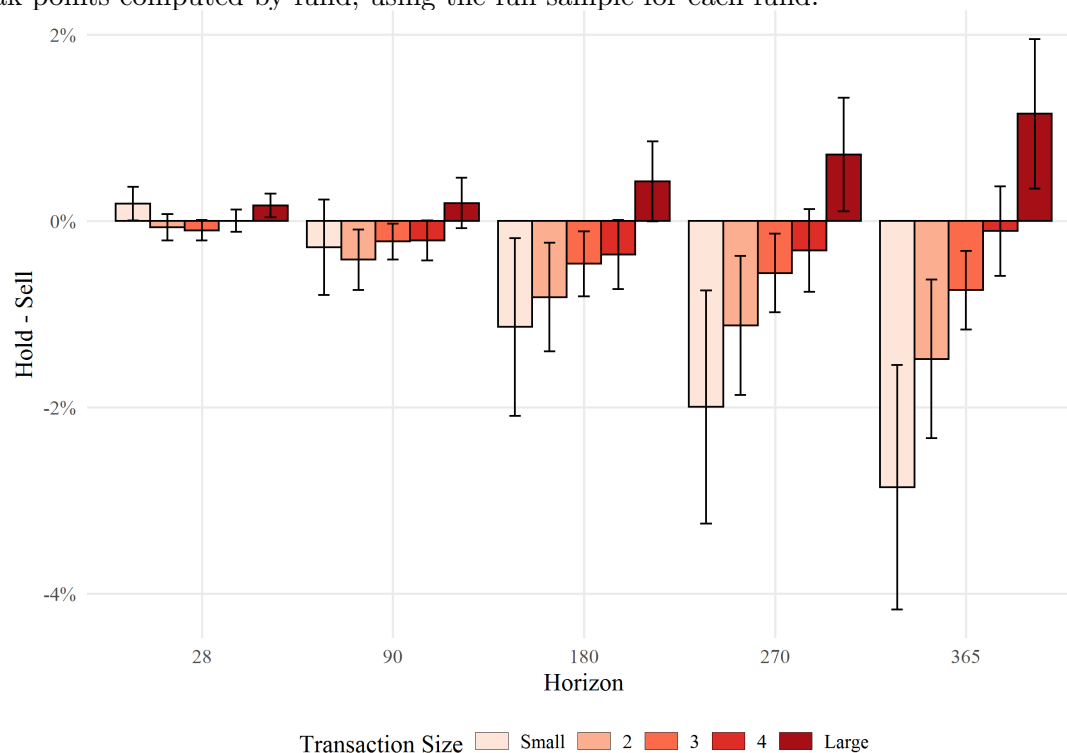
This table presents position-level estimates of a linear probability model (in percentage points) for the likelihood of trading a given stock. The key explanatory variables of interest are indicators corresponding to 20 bins of past benchmark-adjusted returns capped at one year, where the tenth bin is the omitted category. We control for fund characteristics including lagged log(AUM), prior-month turnover, the annual volatility of a funds benchmark-adjusted returns, and prior month loadings on Fama-French Cahart regressions (calculated using the Dimson (1979) procedure using 1 year of prior daily returns). We adjust for position-level characteristics including linear and quadratic terms in holding lengths (overall and since last buy) and position sizes(% AUM) at the beginning of the day. The coefficients and t-statistics are reported for the estimates on each bin.

	Buying Probability Given Buy Day				Selling Probability Given Sell Day			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bin 1	-0.058 (-1.574)	-0.068 (-1.821)	-0.108** (-2.845)	-0.282* (-2.409)	2.476*** (15.105)	2.455*** (14.925)	2.177*** (13.598)	1.382*** (6.484)
Bin 2	0.067* (2.068)	0.065 (1.719)	0.057 (0.697)	0.024 (-1.843)	1.452*** (12.849)	1.438*** (12.627)	1.189*** (10.777)	0.881*** (5.803)
Bin 3 to 5	0.071** (3.005)	0.063* (2.567)	0.047 (1.852)	-0.041 (-0.801)	0.761*** (10.928)	0.752*** (10.666)	0.597*** (9.161)	0.467*** (5.064)
Bin 6 to 9	0.033* (2.245)	0.028 (1.787)	0.021 (1.340)	-0.009 (-0.328)	0.170*** (5.833)	0.166*** (5.477)	0.128*** (4.769)	0.084 (1.934)
Bin 11 to 15	-0.073*** (-4.110)	-0.078*** (-4.143)	-0.057*** (-3.741)	-0.034 (-1.166)	-0.007 (-0.194)	0.010 (0.283)	0.017 (0.526)	0.115** (2.618)
Bin 16 to 18	-0.177*** (-5.816)	-0.185*** (-4.668)	-0.183*** (-6.291)	-0.203** (-3.679)	0.488*** (6.794)	0.500*** (6.831)	0.424*** (6.240)	0.928*** (8.674)
Bin 19	-0.240*** (-6.678)	-0.248*** (-6.705)	-0.255*** (-7.497)	-0.264** (-3.043)	0.971*** (10.039)	0.973*** (9.985)	0.870*** (9.295)	1.448*** (9.784)
Bin 20	-0.300*** (-7.254)	-0.308*** (-7.271)	-0.345*** (-8.432)	-0.304** (-2.679)	2.081*** (15.377)	2.082*** (15.245)	1.916*** (14.446)	2.070*** (10.492)
Fund Control	Yes	Yes	No	Yes	Yes	Yes	No	Yes
Fixed Effects	None	Date	Fund × Date	Stock × Date	None	Date	Fund × Date	Stock × Date
R ²	0.034	0.041	0.285	0.346	0.005	0.009	0.164	0.390
N	29.3M	29.3M	30.2M	22.4M	27.8M	27.8M	28.6M	21.0M

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure A.2. Post-trade returns relative to counterfactual, split by transaction size

This figure presents differences between average factor-neutral returns of stocks bought/sold and those of random buy/sell counterfactual strategies for buy and sell trades, grouped by transaction size. The bracket at the top of each bar is the 95% confidence interval of the point estimate at each horizon. Confidence intervals in brackets are computed using double-clustered standard errors, calculated as described in Appendix A.4. Transaction sizes are computed as a % of a fund's AUM, and broken into quintiles with break points computed by fund, using the full sample for each fund.



attention is stretched thin. Prior work proposes that higher turnover is indicative of better trading opportunities, and thus higher future benchmark adjusted return, finding supporting evidence in mutual fund data (Pástor, Stambaugh, and Taylor 2017). While high turnover may indeed reflect better buying opportunities (buying performance measures are similar with higher gross buying volume), the mechanism outlined in the main text suggests that an asymmetric allocation of attention towards buying decisions would exacerbate the PM's tendency to sell poorly. Table A.10 shows that, consistent with such a mechanism, these periods are associated with particularly poor performance of selling decisions.

In the main text, we present evidence that the variation in heuristic intensity, using

Table A.10. Buying and selling skill for subsamples with different levels of turnover

This table shows the factor neutral buying and selling vs. counterfactuals, but splitting the sample between funds with higher and lower turnover and different periods with higher and lower turnover. There is little difference between buying and selling skill for high and low turnover funds in the cross section, but a substantial deterioration in selling skill relative to the counterfactual in periods of high turnover relative to a fund's typical (median) level.

	Panel A: Buys					Panel B: Sells				
	28	90	180	270	365	28	90	180	270	365
Low Turnover Funds	0.34	0.44	0.58	0.6	0.61	0.06	-0.06	-0.15	-0.33	-0.66
(below median)	(0.06)	(0.13)	(0.21)	(0.28)	(0.37)	(0.05)	(0.10)	(0.16)	(0.22)	(0.27)
High Turnover Funds	0.32	0.63	0.72	0.91	1.08	-0.09	-0.56	-1.25	-1.37	-1.21
(above median)	(0.07)	(0.12)	(0.18)	(0.25)	(0.28)	(0.07)	(0.18)	(0.29)	(0.37)	(0.43)
Low Turnover Periods	0.31	0.47	0.89	1.14	1.25	0.00	-0.04	-0.19	-0.18	-0.13
(below fund median)	(0.06)	(0.12)	(0.19)	(0.25)	(0.33)	(0.06)	(0.13)	(0.21)	(0.30)	(0.32)
High Turnover Periods	0.44	0.59	0.80	0.84	1.18	0.05	-0.28	-0.66	-0.77	-0.90
(above fund median)	(0.07)	(0.15)	(0.24)	(0.32)	(0.40)	(0.07)	(0.14)	(0.19)	(0.25)	(0.28)

measures which sort across time within-manager, is linked with negative selling performance. We find that there is fairly limited variation in between manager differences in heuristics intensity. As one way of seeing this, we find that the overall result that heuristic intensity is related to selling underperformance is not sensitive to whether we compute within-fund sorts or across fund sorts. Table A.11 shows that we get similar results to the main text if we sort PMs into bins based on propensity to sell extremes at each point in time, suggesting that our results are not driven by persistent cross-sectional differences across funds in tendency to use heuristics but instead by time-variation in our heuristics proxy within-manager over time. This provides further evidence consistent with selling underperformance being related to an attentional mechanism rather than a persistent trait. In addition, we find that heuristic intensity is also largely uncorrelated with fund-level characteristics (for example, style or portfolio construction characteristics). These results are presented in A.12, which shows that there is no strong relationship between a fund's average style characteristics, concentration, turnover, or longer term focus and its average level of heuristic intensity.

A.9 Results using alternative currency adjustments

All of our analysis uses prices in USD, which implies a specific perspective on currency hedging. Because our sample has strategies which invest globally and have end clients

Table A.11. Post-trade sell returns relative to counterfactual by heuristics intensity, robustness checks

This table presents average returns relative to random sell counterfactuals for sell portfolios sorted by heuristics intensity, where the heuristics intensity measure is sorted across funds by week (panel I) and within funds over time (panel II). We divide these measures into four bins from Lowest, Low-Med, Med-High and Highest, based on their rankings. Columns represent factor-neutral sell performance measures at the following horizons: 1 month, 3 months, 6 months, 9 months, and 1 year. Heteroskedasticity and autocorrelation robust standard errors, computed using the method described in the section A.4, are reported in parentheses.

Heuristics Intensity (weekly)	Bins	Sell Results, Factor-neutral				
	Horizon	28	90	180	270	365
I. Across Funds, by week	Lowest	-0.07 (0.07)	-0.21 (0.14)	-0.29 (0.20)	-0.42 (0.24)	-0.28 (0.33)
	Low-Med	0.00 (0.06)	-0.07 (0.11)	-0.16 (0.15)	-0.12 (0.21)	0.03 (0.24)
	Med-High	0.05 (0.06)	0.05 (0.14)	-0.13 (0.21)	-0.20 (0.22)	-0.31 (0.26)
	Highest	0.11 (0.09)	-0.59 (0.19)	-1.33 (0.31)	-1.57 (0.39)	-2.20 (0.47)
II. Within Funds, by week	Lowest	-0.10 (0.08)	-0.30 (0.13)	-0.26 (0.18)	-0.30 (0.24)	-0.11 (0.31)
	Low-Med	-0.04 (0.06)	-0.05 (0.10)	-0.27 (0.13)	-0.39 (0.23)	-0.24 (0.26)
	Med-High	0.16 (0.07)	-0.02 (0.15)	-0.30 (0.22)	-0.37 (0.26)	-0.62 (0.29)
	Highest	0.07 (0.09)	-0.50 (0.18)	-1.15 (0.29)	-1.34 (0.41)	-1.90 (0.47)

in several countries other than the United States, this may not account for certain currency risks that managers are accounting for in their analysis.

To test this proposition, we calculate the modal country each strategy is invested in, and convert all of the factor-neutral security returns to that currency. Finally, we calculate the period returns for this portfolio, and average the Hold - Sell and Buy - Hold portfolios across time and strategies in the same manner as the USD based results in the paper. Results of this exercise are presented in table A.13.

Table A.12. Cross-sectional correlations between fund average heuristic intensity measure and portfolio characteristics

This table shows the correlation between average heuristic intensity and average portfolio characteristics. The heuristic intensity (propensity to sell in extreme bins) is averaged over the full fund's history, as are the portfolio characteristics. This shows that correlations are near-zero, indicating that the average use of extreme returns by a fund is largely uncorrelated with a strategy's concentration, style, holding horizon, and turnover.

Characteristic	Cross-sectional Correlation
Concentration	0.02
Percent of Holdings held in Large Blocks	-0.01
Percent Long-Term Holdings	-0.02
Turnover	-0.02
Portfolio Average Book to Market Quintile	-0.05
Portfolio Average Size Quintile	-0.03
Portfolio Average Momentum Quintile	0.02

Table A.13. Post-trade returns relative to counterfactual, factor-neutral, hedged to modal portfolio currency

This table presents the average beta-neutral counterfactual returns for buy and sell trades hedged to each portfolio's modal currency. Standard errors are in parenthesis below the point estimates, calculated by the method described in Appendix A.4.

Horizon	28	90	180	270	365
Buy - Hold	0.24 (0.05)	0.40 (0.09)	0.59 (0.13)	0.57 (0.18)	0.63 (0.21)
Hold - Sell	-0.11 (0.05)	-0.42 (0.11)	-0.74 (0.17)	-1.01 (0.22)	-1.05 (0.26)

A.10 Results from a calendar-time trading strategy

As discussed above, our baseline approach for assessing performance involves computing average returns across portfolios which are adjusted for systematic factors. In our baseline analysis, we double-cluster standard errors using the method described in Appendix A.4, which accounts for autocorrelation in contemporaneous trades of the same fund and across funds. Above, we find that the average stock sold outperforms the average stock held by about 80 bp over a one year horizon with a standard error of about 20 bp. The astute reader will note that such a performance differential would likely be very difficult to detect statistically if PMs were trading on many common systematic factors. As a result, our standard error calculations imply that one should be able to construct a calendar time strategy associated with buying stocks that PMs recently sold and taking short positions in stocks that were recently held which should have a high Sharpe ratio due to strong benefits from diversification. In this section, we will show that this is indeed the case, which provides an additional check on the robustness of our main results and inference.

Before describing our approach, we note our dataset is not ideally suited for this form of analysis. We have an unbalanced panel in which the number of portfolios changes over time, so such an approach implicitly overweights the portfolios which appeared early on in our sample. Second, since PMs in our sample operate in many markets, there could be exposures to risk factors/style tilts that our current analysis differences out which are less easy to address by specifying constant loadings on a common set of risk factors. Ultimately, we believe that the extremely low degree of correlation across funds who trade at the same time (or even between stocks bought/sold by the same PM at the same time) should mitigate precisely these types of concerns that calendar time adjustments would normally be used to assuage. In contrast, we would expect a lot of correlation if they were trading on a set of common aggregate factors, though our estimates will find minimal exposures, in line with the similarity between our baseline factor-neutral and unadjusted performance results in Table 3.

In line with the standard approach of the literature, we conduct our analysis at the monthly frequency. To generate the portfolio, on a monthly basis, we construct signals based on the volume of securities sold and value of securities held on each given day by each given fund. The signal produced by each fund-day is the percent of total dollar volume made up by each security within a trading category. For example, if I

sold \$5 worth of Apple, and overall I sold \$10 worth of securities, the fund/day weight on Apple within the sell signal would be 50%. In other words, for security i at time t in portfolio j and in transaction type $k \in \text{buy, hold, sell}$, we construct a signal as:

$$S_{k,i,j,t} = \frac{\text{DollarVolume}_{k,i,j,t}}{\sum_{i=1}^N \text{DollarVolume}_{k,i,j,t}}$$

where *DollarVolume* is the price times the number of shares traded (or held, in which case we use portfolio weights).

This measure computed within each fund/day for each trade type. Days without sells are dropped from the counterfactual hold signal. The monthly signal for month m becomes

$$S_{k,i,m} = \left(\frac{1}{N_{port}} \right) \left(\frac{1}{N_m} \right) \sum_{j=1}^{N_{port}} \sum_{t=1}^{N_m} S_{k,i,j,t}$$

The short position of each strategy is given by the appropriate “hold” signal, and the long position of each is given by the “sell” signal, with leverage normalized to the equivalent of a 100% long-only weight on each side. In order to make sure that weights have equal leverage on each side, we normalize them in the following way:

$$S_{i,m}^{sell} = I(S_{sell,i,m} > S_{hold,i,m})(S_{sell,i,m} - S_{hold,i,m}),$$

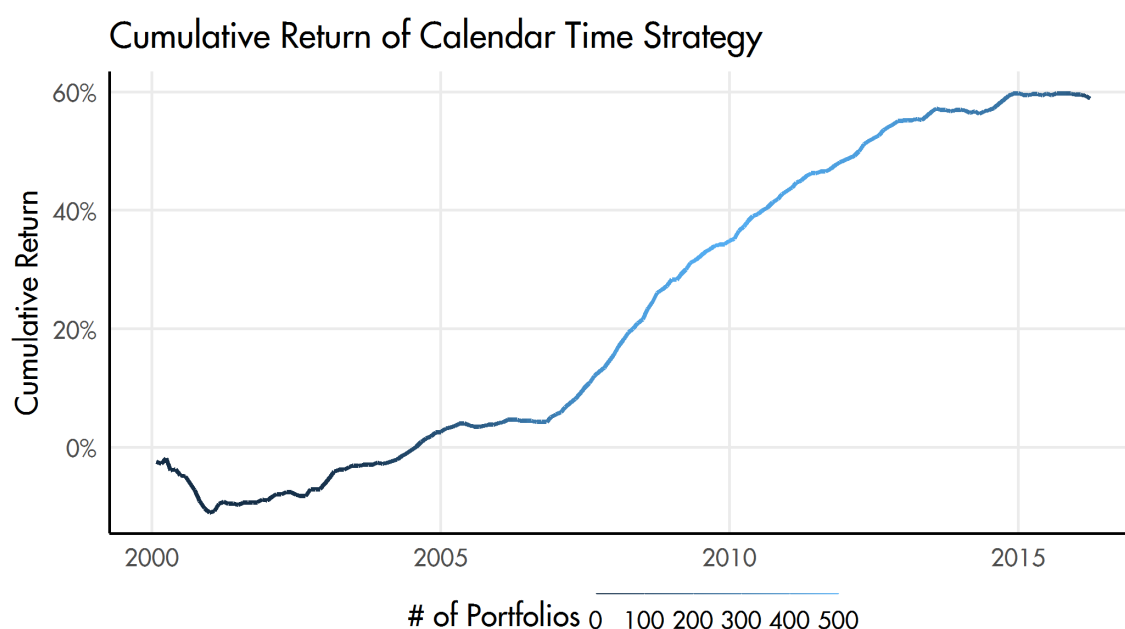
$$S_{i,m}^{hold} = I(S_{hold,i,m} > S_{sell,i,m})(S_{hold,i,m} - S_{sell,i,m})$$

Since our signal predicts long-run underperformance, we construct 12 portfolios associated with the signal month, each at a different lag. We repeat this analysis, constructing portfolios which are associated with stocks sold minus stocks held, for various look-back periods. Specifically, we look at stocks which were traded/held between 1 and 12 months ago to form the portfolios. For example, given the signal in December 2009, we construct a January 2010 portfolio based on that signal, a portfolio which holds the same stocks with the same weights in February 2010, and so on through December.

To get an estimate most directly comparable to the one year cumulative returns discussed in the text, we average returns from the 12 portfolios formed with trading information from the prior 1 through 12 months. Figure A.3 shows the time-series of returns to this long-short strategy. The color of the line also indicates the number of portfolios included in the construction of trading signals at each point in time, which

Figure A.3. Cumulative returns for a trading strategy which purchases stocks recently sold and sells short stocks which were recently held over the prior year

This figure reports the time series of cumulative monthly returns associated with a trading strategy which takes long positions in stocks which were disproportionately sold over the prior year and short positions in stocks which were recently held over the same periods. Note that our sign convention differs from the rest of the paper, so positive numbers indicate that sells outperform stocks held (which acts as a drag on performance). Color of the line indexes the number of strategies we observe in a given month over the course of our full sample period. Please see the text for further details on portfolio construction.



also shows one reason why we do not view this as the most appropriate way to analyze the data. As noted above, an important limitation in this analysis is that we have a somewhat unbalanced panel of strategy observations. We then average the twelve Sell - Hold portfolios, yielding a diversified portfolio associated with 1 year worth of past signals.

As is clear from the graph, the strategy does quite well. It has both positive returns and low volatility, and improves as we see a larger number of portfolios enter our sample. Reassuringly, the performance of the strategy is best during periods of time when our signals are aggregated over the largest number of funds.

As is clear from the steep slope and low volatility of the graph, this portfolio

Table A.14. Sharpe Ratios for a variety calendar time portfolios which purchase stocks recently sold and sell short stocks which were recently held

This table reports annualized Sharpe ratios associated with a number of long-short strategies based on a signal constructed from the previous sells and holds of the managers in our sample. In the first row, we report raw Sharpe ratios. In subsequent rows, we report Sharpe ratios that obtain after hedging exposures to different factors. In each case, we average over multiple portfolios which are formed based on trading activity from one or more month ago. Columns indicate which periods are averaged when computing the overall portfolio. Signals are constructed based on the percent of daily volume a security made up in the sell portfolio, and based on the portfolio weight relative to other holds for the hold portfolio (please see text for further details on portfolio construction). Sharpe ratios are computed assuming that stocks recently sold are purchased for the long end and the stocks recently held are sold short.

	1 Year	Months 2:12	Months 1:6	Months 7:12
No Adjustment	1.496	1.312	1.037	1.150
CAPM	1.484	1.296	1.030	1.138
Three Factor	1.669	1.469	1.141	1.260
Four Factor	1.671	1.467	1.130	1.276

has a very large Sharpe ratio, with both low volatility and high returns (though the numerator is largely a function of how we normalize the weights). Table A.14 computes the Sharpe ratio associated with our one year portfolio. In order to make sure that different systematic factors are not driving the results, we adjust for a variety of factor models: the CAPM, the Fama-French 3 factor model, and the Carhart 4 factor model and report Sharpe ratios associated with portfolios which are hedged with respect to these factors.

We also perform the same analysis with different formation windows, to show that the specific choice of one year is not driving the results. The portfolios based on the first 6 months after trade have almost the same Sharpe ratios as those based on the last 6 months. In addition, we exclude the first month due to concerns about trading costs, and demonstrate that the portfolio does about as well without include the first month following each “sell” observation. The relationship between the selling signal and future performance is fairly persistent, so the lower denominator tends to dominate in the Sharpe ratio calculations, leading them to increase as we aggregate over longer periods.

Table A.15. Abnormal returns and factor loadings for trading strategy which purchases stocks recently sold and sells short stocks which were recently held over the prior year

This table reports annualized Sharpe ratios, along with monthly alphas and factor loadings associated with a trading strategy which takes long positions in stocks which were disproportionately sold over the prior year and short positions in stocks which were recently held over the same periods. We report measures for monthly performance for a number of factor specifications. Robust standard errors are in parentheses. Please see the text for further details on portfolio construction.

	Ann. Sharpe	Monthly Alpha (%)	Mkt - RF	SMB	HML	Mom
No Adjustment	1.496	0.238 (0.040)				
CAPM	1.484	0.236 (0.0418)	0.00702 (0.011)			
Three Factor	1.669	0.257 (0.0383)	0.0128 (0.0113)	-0.0377 (0.0167)	-0.0342 (0.0187)	
Four Factor	1.671	0.257 (0.0389)	0.0124 (0.0127)	-0.0375 (0.0165)	-0.0344 (0.0181)	-0.000769 (0.0104)

Table A.15 reports alphas and factor loadings for these portfolios, the latter of which are almost always close to zero, indicating that this signal is largely unrelated to potential known risk factors which are related to returns. These results underscore our findings from above suggesting that PMs in our sample appear to be trading on individual stocks rather than seeking systematic exposures to common factors.

In closing, we note that actually implementing this trading strategy would likely result in lower returns than we see here because of the high transaction associated with a high turnover strategy investing in large numbers of securities globally. It is also not likely to be tradeable in practice because of the need for daily holdings information across a variety of skilled managers. As such, we view this exercise as largely supporting our baseline result and providing another perspective on the robustness of our standard error calculations.